Data Analytics Capstone

Cluster-Adaptive Prompting for Multi-Turn Travel and Tourism Conversational Systems

Final Report

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Abstract

Online travel agencies (OTAs) deploy large language model (LLM) agents for package booking, yet static prompts ignore user diversity while unguarded few-shot prompts risk entity leakage and rule violations. This study tests a neuro-symbolic system joining power analysis, clustering, deep learning embeddings, and agentic prompting with rule checks. The study proceeds across four stages formulated as separate research questions, **RQ1-Identify, RQ2-Personalize, RQ3-Govern, and RQ4-Optimize**, where each stage proves a prerequisite, drops non-viable methods, and transfers its validated causal artifacts to the next stage. First, the **RQ1-Identify** stage proves shopper behavior can be segmented from opening prefixes without outcome confounding, dropping noise-rejecting and whole-dialogue models to transfer three archetype centroids, vector sub-indices, and a prefix router to the **RQ2-Personalize** stage. Second, the **RQ2-Personalize** stage uses those routed labels and exemplar pools to prove cluster-adaptive few-shot prompting aligns assistant style and word-length pacing to the persona without semantic drift, dropping static prompts and transferring the adaptive generator to the **RQ3-Govern** stage. Third, the **RQ3-Govern** stage tests the adaptive generator under policy stress to prove pairing symbolic rule checks with slot masking eliminates entity leakage and enforces booking rules while preserving style, dropping unguarded prompts and transferring the governed agent to the **RQ4-Optimize** stage. Fourth, the **RQ4-Optimize** stage deploys the cascaded **RQ1-Identify** router, **RQ2-Personalize** adaptive prompts, and **RQ3-Govern** rule-governed slot ledger in closed-loop multi-turn simulations to prove that matching memory tracking and trade-off framing to each archetype compresses turns and removes repeat slot asks. Multi-turn user–agent dialogues form the unit of analysis across $N = 1{,}334$ Microsoft Frames dialogues (2016–2017; 70% golden $n = 933$, 30% test $n = 401$) and $N = 1{,}499$ MultiWOZ 2.2 dialogues (2020) over 10 weeks in 2026,

built in Python 3.11 (*scikit-learn*, *Statsmodels*, Facebook AI Similarity Search [*FAISS*], Sentence-BERT [SBERT]) on an 8-core Apple M-series CPU and cloud GPU/TPU via Gemini 2.5 Flash and Pro ($\alpha = .05$). Across the four stages, empirical findings confirm each handoff: in the **RQ1-Identify** stage, density clustering dropped 24.4%–39.1% of sessions as noise and whole-dialogue clusters suffered outcome confounding ($\kappa = .24$); dropping both, 100%-coverage *K*-Means ($k = 3$) on a 46-d *Blocks* prefix matrix yielded three stable archetypes (Adjusted Rand Index = .89), namely *Decisive Booker*, *Open Explorer*, and *Rigid-Constraint Seeker*, routing test prefixes at 70.1% accuracy ($\kappa = .535$, area under curve [AUC] = .871) at five messages and 97.3% ($\kappa = .958$) at nine to hand off the router and FAISS sub-indices to the **RQ2-Personalize** stage. In the **RQ2-Personalize** stage ($N = 33$), coarse unrefined clusters failed ($p = .211$) and were dropped, whereas the carried-forward refined router and slot-masked FAISS exemplars raised SBERT style cosine from .453 (static) and .530 (zero-shot) to .597 ($p < .001$) and cut pacing error from 10.12 to 3.87 words ($p = .007$), handing off the adaptive generator to the **RQ3-Govern** stage. In the **RQ3-Govern** stage (2×2, $N = 144$), unguarded few-shot prompts leaked exemplar entities (.359 rate) and caught only 5.6% of violations; dropping unguarded prompts and activating YAML rules with *[SLOT]* masking drove leakage to .000 ($p < .001$), lifted violation sensitivity to 98.6% ($p < .001$), and raised Slot F_1 to .952 while preserving style, handing off the governed agent to the **RQ4-Optimize** stage. In the **RQ4-Optimize** stage (2×3, $N = 180$), cascading all upstream stages with a slot ledger and persona trade-offs removed repeat asks (.109 → .000, $p < .001$) and cut turns by 29.5%–42.7% ($p < .001$; 8.73 → 5.00 turns for *Rigid-Constraint Seekers*), operationalizing turn economy in a FastAPI/Redis service.

Keywords: conversational artificial intelligence, customer archetypes, large language models, in-context learning, feature importance, business-rule enforcement, dialogue turn economy

Introduction

Background and Context

Conversational artificial intelligence (AI) assistants have become a primary digital service channel in travel, tourism, and hospitality, supporting destination discovery, multi-package comparison, and itinerary booking across online travel agencies (OTAs), tour operators, and hotel contact centers (Buhalis & Sinarta, 2019; Ivanov & Webster, 2019; Jannach et al., 2021). Unlike single-turn informational queries, vacation package booking is a consultative, multi-turn negotiation in which travelers balance competing constraints across origin city, destination city, departure and return dates, trip duration, party composition, hotel star rating, and total budget (El Asri et al., 2017). Empirical observations of travel agency dialogues show that shoppers exhibit markedly different behavioral trajectories: some state complete trip parameters up front and seek immediate confirmation, others explore multiple destinations and hotel tiers side by side, and others maintain rigid date and luxury requirements that trigger constraint deadlocks when preferred inventory exceeds their budget ceiling (Inaba et al., 2022; Melián-González et al., 2021).

Despite this behavioral heterogeneity, commercial large language model (LLM) travel assistants typically operate under a single static system prompt and a uniform slot-filling questionnaire. Sequentially querying one missing slot per turn or presenting a single hotel option without trade-off framing inflates conversation length, triggers repetitive slot re-asking, and drives customer abandonment before quotation agreement is reached. For hospitality operators and OTA engineering teams, excess conversational turns also inflate cloud LLM inference latency and token expenditure while degrading self-service conversion rates (Buhalis & Sinarta, 2019; Melián-González et al., 2021).

Conversely, attempting to personalize an LLM assistant purely through dynamic few-shot in-context learning (ICL) introduces two severe operational hazards (Brown et al., 2020; Liu et al., 2022; Rebedea et al., 2023). First, injecting unmasked historical dialogue exemplars into the prompt induces factual entity leakage, in which the generative model copies stale hotel names, prices, dates, or promotional codes from retrieved examples into the active customer session. Second, stylistic prompt instructions compete for model attention with enterprise booking policies, causing unguarded LLMs to overlook mandatory visa lead-time windows, passport validity rules, group occupancy thresholds, discount ceilings, or minimum package budgets (Yao et al., 2024). Resolving the tension between stylistic adaptability, policy compliance, and turn economy therefore requires a neuro-symbolic architecture that couples early-turn behavioral routing with deterministic symbolic verification.

Problem Statement

The objective of this study is to model, predict, and optimize six quantitative dependent variables (Y), specifically unsupervised cluster separation Silhouette and Davies–Bouldin index (Y_1), de-lexicalized Sentence-Bidirectional Encoder Representations from Transformers (SBERT) style convergence cosine and archetype pacing calibration error in words (Y_2), policy rule adherence rate decomposed into violation enforcement sensitivity and compliant acceptance specificity (Y_3), exemplar entity leakage rate (Y_4), factual slot coverage F_1 (Y_5), and natural-log-transformed turns to goal completion (TGC) and redundant elicitation rate (RER; Y_6), for the unit of analysis of individual multi-turn customer–assistant travel dialogue trajectories and turn-level agent responses using four manipulated independent variables (X): clustering algorithm and prefix feature representation (X_1, research question 1 [RQ1]), in-context prompting strategy (X_2, RQ2), declarative rule-engine governance mode crossed with prompting strategy (X_3, RQ3), and

prompting strategy crossed with refined user archetype (X_4, RQ4) over the archival 2016–2017 Microsoft Frames corpus (N = 1,334 analytic dialogues) and 2020 MultiWOZ 2.2 benchmark (N = 1,499 analytic dialogues) evaluated during the 2026 experimental window. Success is evaluated at the $\alpha = .05$ significance level using analysis of variance (ANOVA) F-tests, partial eta-squared (η_p^2) effect sizes, bootstrap Adjusted Rand Index (ARI $\geq .60$), held-out Cohen's κ and multi-class ROC-AUC, dual-judge inter-rater κ, and percentage reductions in conversational turns and redundant slot requests.

Purpose of the Study

The purpose of this quantitative experimental study is fourfold: (1) to segment multi-turn travel shoppers into stable, interpretable behavioral archetypes using opening-prefix embeddings and structured LLM behavioral ratings, while quantifying feature importance to construct an accurate early-turn prefix classifier; (2) to evaluate whether routing test prefixes into archetype-partitioned Facebook AI Similarity Search (FAISS) exemplar sub-indices and pacing directives improves discourse style convergence and word-length calibration; (3) to test in a 2 × 2 factorial design how a declarative YAML Ain't Markup Language (YAML) rule engine and slot masking interact with cluster-adaptive prompting to govern policy compliance, factual slot grounding, and entity leakage; and (4) to quantify in a 2 × 3 closed-loop simulation how pairing a persistent informed-slot ledger with archetype-tailored trade-off policies compresses turns to package agreement across distinct traveler personas.

Research Problems and Research Questions

To operationalize the problem statement across the four-stage **RQ1-Identify → RQ2-Personalize → RQ3-Govern → RQ4-Optimize** architecture, the study investigates four linked research questions:

- **Research Question 1 (RQ1-Identify Stage):** Which unsupervised clustering algorithm (*K*-Means, Ward agglomerative clustering, or Hierarchical Density-Based Spatial Clustering of Applications with Noise [HDBSCAN]) best separates multi-turn travel dialogue representations, which semantic and behavioral features carry the highest discriminative importance, and how accurately can the resulting refined user archetypes be routed from early conversation prefixes ($M \in \{3, 5, 7, 9\}$ user messages)?
- **Research Question 2 (RQ2-Personalize Stage):** Does dynamic cluster-adaptive few-shot prompting, combining archetype-partitioned, slot-masked FAISS exemplar retrieval with routed archetype pacing directives, significantly increase de-lexicalized SBERT style convergence and reduce archetype pacing calibration error relative to zero-shot and static few-shot baselines?
- **Research Question 3 (RQ3-Govern Stage):** In a 2 × 2 factorial design crossing declarative rule-engine mode (*Rules Off* vs. *Rules Active*) with prompting strategy (*Zero-Shot* vs. *Cluster-Adaptive Few-Shot*), what are the main and interaction effects on policy rule adherence (violation sensitivity and compliant specificity), exemplar entity leakage rate (ELR), factual slot coverage F_1, and style alignment?
- **Research Question 4 (RQ4-Optimize Stage):** In a 2 × 3 factorial multi-turn simulation crossing prompting policy (*Zero-Shot* vs. *Cluster-Adaptive Few-Shot* with a *Verified Informed Slot Ledger*) and refined user archetype (*Decisive Booker*, *Open Explorer*, *Rigid-Constraint Seeker*), what are the main and interaction effects on log-transformed turns to goal completion (TGC) and redundant elicitation rate (RER)?

Table 1 maps each research question to its formal null (H_0) and alternative (H_a) hypotheses, experimental design, and statistical testing framework.

Table 1

Research Questions, Null and Alternative Hypotheses, and Analytical Framework

RQ	Research question	Null and alternative hypotheses	Experimental design and statistical test
RQ1	Which clustering algorithm best separates dialogue embeddings, which features drive separation, and which early-routable archetypes emerge?	H_{01}: $\mu_{KM} = \mu_{Ward} = \mu_{HDB}$ on Silhouette; prefix routing $\kappa \leq 0$. H_{a1}: At least one algorithm mean differs, and hybrid prefix features yield stable (ARI $\geq .60$), routable archetypes ($\kappa > 0$).	Resampling benchmark ($N = 159$ runs across 53 matched $n = 500$ seeds: one-way ANOVA, matched RM-ANOVA, Friedman, Welch, Kruskal–Wallis) + feature importance (η^2, $\Delta\kappa$) + Train/CV/Test prefix routing
RQ2	Does cluster-adaptive few-shot prompting improve style convergence and pacing calibration over zero-shot and static few-shot baselines?	H_{02}: $\mu_{ZS} = \mu_{Static} = \mu_{Adaptive}$ on style cosine and pacing error. H_{a2}: $\mu_{Adaptive} > \mu_{Static}, \mu_{ZS}$ on style cosine and $\mu_{Adaptive} < \mu_{Static}, \mu_{ZS}$ on pacing error.	One-way repeated-measures ANOVA (RM-ANOVA; $N = 33$ held-out dialogues × 3 conditions = 99 generations; Greenhouse–Geisser correction, Friedman test, Holm-adjusted paired t contrasts)
RQ3	How do a declarative rule engine and cluster-adaptive prompting interact to govern rule adherence, entity leakage, slot F_1, and style alignment?	H_{03}: $\alpha_{Rules} = \beta_{Prompt} = (\alpha\beta)_{Rules \times Prompt} = 0$ across RAR, ELR, Slot F_1, and style cosine. H_{a3}: Active rules raise RAR and Slot F_1 and interact with prompting to eliminate few-shot entity leakage.	2 × 2 factorial ANOVA ($N = 144$ evaluations across 36 stratified held-out dialogues, $n = 36$/cell; ordinary least squares [OLS], heteroscedasticity-consistent [HC3], Aligned Rank Transform [ART], sensitivity/specificity split; dual-LLM judge κ)
RQ4	Does cluster-adaptive prompting with an informed-slot ledger reduce turns to goal completion and redundant elicitation across user archetypes?	H_{04}: $\alpha_{Prompt} = \beta_{Arch} = (\alpha\beta)_{Prompt \times Arch} = 0$ on ln(TGC) and RER. H_{a4}: Cluster-adaptive prompting reduces ln(TGC) and RER with a significant prompting × archetype interaction.	2 × 3 between-subjects factorial ANOVA on ln(TGC) and RER ($N = 180$ unique held-out seed dialogues, $n = 30$/cell; OLS, HC3, ART, Welch t, Mann–Whitney U; blind persona κ)

Note. RQ = research question; KM = K-Means; HDB = HDBSCAN; ARI = adjusted Rand index; RM-ANOVA = repeated-measures analysis of variance; RAR = rule adherence rate; ELR = exemplar leakage rate; TGC = turns to goal completion; RER = redundant elicitation rate. All hypothesis tests are evaluated at $\alpha = .05$.

Contributions and Expected Value

Practical contribution. This study delivers a deployable neuro-symbolic conversational blueprint that eliminates the 35.9% exemplar entity leakage rate (55.6% session incidence) of

unguarded few-shot prompting while raising policy-violation enforcement sensitivity from 5.6% to 98.6% and compressing multi-turn package negotiations by 29.5% to 42.7% without redundant slot questions.

Technical contribution. Methodologically, the study (1) proves that truncating dialogues to opening user prefixes ($M = 9$) and fusing 20 Gemini semantic principal components with 26 encoded LLM behavioral columns (*Blocks*) resolves end-of-dialogue outcome confounding and yields high bootstrap stability (ARI = .89/.87); (2) establishes a three-part feature-importance methodology (cluster variance explained η^2, standardized logistic regression coefficient norms, and held-out permutation drop $\Delta\kappa$) that directly governs feature selection, centroid naming, and prefix routing (70.1% accuracy, $\kappa = .535$, ROC-AUC = .871 at $M = 5$); and (3) provides rigorous parametric (ordinary least squares [OLS], Greenhouse–Geisser), heteroscedasticity-consistent (Welch, MacKinnon–White HC3), and non-parametric (Kruskal–Wallis, Friedman, Mann–Whitney *U*, Aligned Rank Transform [ART]) triangulation across all four stages.

Value to stakeholders and target users. For travelers, the system provides concise, non-repetitive consultations tailored to their decision style: rapid confirmation for *Decisive Bookers*, side-by-side package comparisons for *Open Explorers*, and immediate budget-feasible trade-offs for *Rigid-Constraint Seekers*. For OTA engineering teams, tour operators, and compliance officers, the architecture reduces LLM inference compute by roughly one-third while enforcing deterministic visa, passport, occupancy, discount, and minimum-budget rules prior to response delivery.

Literature Review

Literature Review Approach

A systematic search of peer-reviewed literature was conducted across the Association for Computational Linguistics (ACL) Anthology, Association for Computing Machinery (ACM) Digital Library, Institute of Electrical and Electronics Engineers (IEEE) Xplore, ScienceDirect, Emerald Insight, and Google Scholar using Boolean combinations of *travel and hospitality chatbots*, *conversational recommender systems*, *task-oriented dialogue corpora*, *deep learning sentence embeddings*, *unsupervised trajectory clustering*, *in-context exemplar retrieval*, and *agentic LLM guardrails*. Studies were included if they provided empirical datasets, mathematical clustering or embedding methods, agentic LLM prompting or guardrail architectures, or statistical power and evaluation frameworks applicable to multi-turn dialogue. Thirty relevant sources, comprising 28 peer-reviewed journal articles, conference proceedings, and reference monographs alongside two explicitly labeled technical preprints, were synthesized to establish the theoretical and methodological foundation for RQ1 through RQ4.

Summary of Key Literature

Table 2 presents the literature relevance matrix across fourteen foundational entries, detailing each study's author and year, domain context, dataset or experimental setting, analytical methods, key empirical findings, and the exact operational inspiration derived for each stage of the **RQ1-Identify → RQ2-Personalize → RQ3-Govern → RQ4-Optimize** research framework.

Table 2

Literature Relevance Matrix Across Travel, Deep Learning, Clustering, and Agentic LLM Domains

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	RQ Linkage & Stage Operationalization
Buhalis & Sinarta (2019); Ivanov & Webster (2019)	Tourism & hospitality service automation	Enterprise hospitality & travel service ecosystems	Real-time 'nowness' co-creation & AI economic framework	Self-service AI creates ROI only when resolving complex inquiries rapidly without human escalation	**RQ4-Optimize Stage & Overarching Goal:** Inspires operationalizing turn economy (TGC) and policy compliance as the end-to-end business target
Melián-González et al. (2021)	Consumer travel chatbot adoption	Survey of *N* = 1,476 travel & tourism consumers	Structural equation modeling of behavioral intention	Perceived task fit and low interaction effort are primary drivers of travel chatbot adoption	**RQ2-Personalize → RQ4-Optimize Stages:** Inspires operationalizing persona pacing calibration and zero redundant slot asks (RER)
Jannach et al. (2021); Inaba et al. (2022)	Conversational recommender systems (CRS)	100+ CRS studies & multi-turn travel agency corpus	Taxonomy of preference elicitation & dialogue act analysis	Travelers differ sharply in exploration breadth and constraint rigidity; static policies cause churn	**RQ1-Identify → RQ4-Optimize Stages:** Inspires operationalizing the 3-archetype taxonomy and persona trade-off policies
El Asri et al. (2017)	Goal-oriented travel package dialogue	Microsoft Frames (1,369 human Wizard-of-Oz dialogues)	Multi-frame slot tracking & frame-switching annotation	Captures nonlinear budget trade-offs, slot revisions, and multi-offer vacation package comparisons	**All Four Stages (RQ1-Identify to RQ4-Optimize):** Inspires operationalizing multi-offer package negotiation on the primary corpus (*N* = 1,334; 933 gold / 401 test)

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	RQ Linkage & Stage Operationalization
Budzianowski et al. (2018); Zang et al. (2020)	Multi-domain task-oriented dialogue	MultiWOZ & MultiWOZ 2.2 (10,438 dialogues across 7 domains)	Corrected turn-level slot state tracking benchmarks	Standardizes slot annotations for hotel, train, attraction, and taxi bookings without airfare packages	**RQ1-Identify Stage:** Operationalizes external cross-corpus domain-shift validation (N = 1,499) before locking Frames centroids
Reimers & Gurevych (2019); Lee et al. (2025)	Deep learning semantic embeddings	SNLI, STS, & multilingual retrieval benchmarks	Siamese BERT (*all-MiniLM-L6-v2*, 384-d) & *gemini-embedding-001* (3,072-d)	Contrastive transformer embeddings map semantically equivalent utterances into cosine space	**RQ1-Identify → RQ2-Personalize Stages:** Operationalizes 3,072-d PCA-20 prefix embeddings in the RQ1-Identify stage and de-lexicalized SBERT style cosine in the RQ2-Personalize stage
Rousseeuw (1987); Davies & Bouldin (1979); McInnes et al. (2017, 2018)	Unsupervised clustering & manifold learning	High-dimensional metric & density clustering benchmarks	Cosine Silhouette, Davies–Bouldin index, HDBSCAN density clustering, & UMAP projection	Silhouette and DB quantify partition compactness; HDBSCAN isolates dense cores but rejects noise	**RQ1-Identify Stage:** Operationalizes 159-run algorithm selection, proving why noise-dropping HDBSCAN is dropped and 100%-coverage K-Means (k=3) is carried to the RQ2-Personalize stage
Hennig (2007); Hubert & Arabie (1985); Pedregosa et al. (2011)	Cluster stability & machine learning	Bootstrap resampling & *scikit-learn* ML suite	Chance-corrected Adjusted Rand Index (ARI) across 80% subsamples & L2 logistic classification	ARI ≥ .60 confirms reproducible structure; regularized linear routers generalize on compact features	**RQ1-Identify → RQ2-Personalize Stages:** Operationalizes bootstrap stability (ARI = .89) and the L2 logistic prefix router handed off to the RQ2-Personalize stage

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	RQ Linkage & Stage Operationalization
Gilardi et al. (2023)	LLM behavioral text annotation	Political & social science coding benchmarks	Zero-temperature LLM structured codebook annotation vs. human crowd workers	Zero-temperature LLMs achieve higher inter-coder reliability than crowd workers on structured scales	**RQ1-Identify & RQ4-Optimize Stages:** Operationalizes the 16-field temperature-0 LLM behavioral codebook (κ_w = .87–.96) fused with prefix embeddings
Brown et al. (2020); Liu et al. (2022); Johnson et al. (2021)	Deep learning & agentic in-context learning	GPT-3 few-shot NLP benchmarks & billion-scale vector search	Few-shot prompting, KATE *k*-NN exemplar selection, & FAISS *IndexFlatIP* search	Semantically matched in-context examples outperform static prompts in sub-millisecond vector lookup	**RQ2-Personalize Stage:** Inspires operationalizing three archetype-partitioned FAISS sub-indices (*k* = 3) handed off to the RQ3-Govern stage
Rebedea et al. (2023); Yao et al. (2024)	Agentic LLM guardrails & policy safety	NeMo Guardrails & τ-bench airline/retail policy suite	Programmable symbolic rails & multi-turn tool-agent policy compliance evaluation	Prompt-only LLM agents violate domain rules in > 40% of complex multi-turn interactions	**RQ3-Govern Stage:** Inspires operationalizing deterministic YAML rule checks and [SLOT] masking on top of RQ2-Personalize adaptive prompts before the RQ4-Optimize stage
Acikgoz et al. (2025); Zhang et al. (2020)	Dialogue & neural generation evaluation	TD-EVAL multi-turn benchmarks & WMT generation tasks	Turn-level factual precision + dialogue comparisons & contextual BERTScore F_1	Multi-turn evaluation requires pairing contextual similarity with fine-grained factual slot audits	**RQ2-Personalize → RQ4-Optimize Stages:** Operationalizes multi-dimensional stage gatekeeping (SBERT style, BERTScore, Slot F_1, ELR, RER)

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	RQ Linkage & Stage Operationalization
Cohen (1988); Faul et al. (2007); Greenhouse & Geisser (1959)	Statistical power & repeated-measures design	Behavioral & experimental design methodology	Noncentral-*F* sample size derivation (G*Power) & sphericity-corrected RM-ANOVA	Noncentral *F* ensures $1 - \beta \geq .80$ at $\alpha = .05$; ε correction prevents Type I error inflation	**All Four Stages (RQ1-Identify to RQ4-Optimize):** Operationalizes a priori sample size power (Table 8) and Greenhouse–Geisser RM-ANOVA in the RQ1-Identify and RQ2-Personalize stages
Seabold & Perktold (2010); Wobbrock et al. (2011)	Econometric & non-parametric factorial ANOVA	Python *Statsmodels* suite & skewed factorial data	OLS Type II/III ANOVA, White/MacKinn on HC3 robust covariance, & Aligned Rank Transform (ART)	HC3 and rank-based ART preserve valid factorial inference under variance heterogeneity and floor bounds	**RQ3-Govern → RQ4-Optimize Stages:** Operationalizes parametric, HC3, and ART factorial validation for the 2×2 and 2×3 cascades

Note. CRS = conversational recommender system; ARI = adjusted Rand index; FAISS = Facebook AI Similarity Search; RM-ANOVA = repeated-measures ANOVA; HC3 = heteroscedasticity-consistent standard error; ART = aligned rank transform.

Thematic Synthesis and Research Gaps

Operationalizing the overarching framework and the RQ4-Optimize stage from hospitality and CRS literature. The four-stage **RQ1-Identify → RQ2-Personalize → RQ3-Govern → RQ4-Optimize** progression was derived directly by synthesizing operational gaps across four distinct literature streams and translating their theoretical constructs into an integrated LLM-and-clustering pipeline. In digital travel and hospitality commerce, competitive differentiation relies on real-time itinerary co-creation ('nowness' service; Buhalis & Sinarta, 2019). Ivanov and Webster (2019) established that automated conversational agents generate net economic value only when they resolve consultative inquiries rapidly without human escalation, while Melián-González et al. (2021) demonstrated across 1,476 consumers that travel chatbot

adoption is governed by perceived task fit and low interaction effort. Corpus studies of conversational recommender systems (CRS) further revealed that human travel agents achieve efficiency by adapting their turn pacing and recommendation strategy to three recurring shopper postures: confirming bookings concisely for decisive buyers, curating side-by-side comparisons for exploratory shoppers, and negotiating date or hotel-tier trade-offs for constraint-bound travelers (Inaba et al., 2022; Jannach et al., 2021). These hospitality and CRS studies provided the direct inspiration for the **RQ4-Optimize stage**: operationalizing business optimization through *Turns to Goal Completion (TGC)*, *Redundant Elicitation Rate (RER)*, a persistent *Verified Informed Slot Ledger*, and archetype-tailored trade-off framing. However, because prior tourism studies relied on post-hoc consumer surveys or manual dialogue-act coding, operationalizing this vision in a live LLM assistant required first building an algorithmic mechanism to identify customer archetypes early in an active session (**RQ1-Identify stage**), personalize generation style (**RQ2-Personalize stage**), and enforce enterprise booking rules (**RQ3-Govern stage**).

Operationalizing the RQ1-Identify stage by fusing deep learning embeddings, clustering theory, and LLM behavioral coding. To operationalize early-turn archetype identification for the **RQ1-Identify stage**, this study drew inspiration from task-oriented dialogue corpora, dense transformer encoders, unsupervised clustering theory, and zero-temperature LLM annotation. While MultiWOZ (Budzianowski et al., 2018) and MultiWOZ 2.2 (Zang et al., 2020) standardize single-service slot tracking, Microsoft Frames (El Asri et al., 2017) uniquely captures multi-turn flight-and-hotel package negotiations under budget ceilings and frame switches, inspiring its selection as the primary discovery and simulation corpus alongside MultiWOZ 2.2 for cross-corpus transfer testing. Siamese Sentence-BERT encoders (Reimers & Gurevych, 2019) and

3,072-dimensional Gemini embeddings (Lee et al., 2025) inspired mapping variable-length dialogue prefixes into cosine space, while clustering validation theory, including Cosine Silhouette (Rousseeuw, 1987), Davies–Bouldin scatter ratios (Davies & Bouldin, 1979), HDBSCAN density estimation (McInnes et al., 2017), UMAP manifold projection (McInnes et al., 2018; Pedregosa et al., 2011), and bootstrap ARI stability (Hennig, 2007; Hubert & Arabie, 1985), inspired the 159-run algorithm benchmark and the 100%-coverage selection gate that drops noise-rejecting HDBSCAN in favor of full-coverage *K*-Means. Crucially, because embedding complete transcripts confounds opening shopper behavior with end-of-session outcomes, Gilardi et al.'s (2023) finding that zero-temperature LLMs outperform human crowd workers on structured codebooks inspired operationalizing the 16-field temperature-0 LLM behavioral schema on opening user prefixes ($M \leq 9$) and fusing its 13 non-outcome behavioral fields (26 encoded columns) with 20 Gemini prefix principal components into the 46-dimensional *Blocks* representation, using feature importance (η^2, $\|\boldsymbol{\beta}\|_2$, $\Delta\kappa$) to prune outcome-confounded fields and train the L2 logistic prefix router handed off to the **RQ2-Personalize stage**.

Operationalizing the RQ2-Personalize and RQ3-Govern stages from in-context learning and neuro-symbolic guardrail literature. Once the **RQ1-Identify stage** hands off a live prefix router and three archetype partitions, the **RQ2-Personalize stage** draws direct operational inspiration from Brown et al.'s (2020) few-shot in-context learning paradigm, Liu et al.'s (2022) KATE nearest-neighbor exemplar selection, and Johnson et al.'s (2021) FAISS inner-product vector indexing: rather than querying a single monolithic example pool (Static Few-Shot), the **RQ2-Personalize stage** operationalizes three archetype-partitioned *faiss.IndexFlatIP* sub-indices paired with routed persona pacing directives so that a shopper classified at message 5 receives

exemplars and word-length norms matched to their exact behavioral archetype. However, multi-turn agent benchmarks by Yao et al. (2024; τ-bench) and Acikgoz et al. (2025; TD-EVAL) demonstrated that prompt-only LLM agents violate domain policies in over 40% of sessions and copy stale entities from retrieved examples. Those failure modes, combined with Rebedea et al.'s (2023) NeMo Guardrails architecture, directly inspired operationalizing the **RQ3-Govern stage**: decoupling probabilistic style prompting from a deterministic YAML rule engine and regex *[SLOT]* exemplar masking in a 2 × 2 factorial design to eliminate entity leakage and enforce 98.6% of policy violations before handing the governed agent to the **RQ4-Optimize stage**.

Operationalizing cross-stage statistical gatekeeping and cascading synthesis. Finally, to ensure that the four stages function as a rigorous causal cascade rather than disconnected experiments, the statistical evaluation literature inspired strict quantitative handoff gates at every stage boundary. Contextual similarity metrics (BERTScore; Zhang et al., 2020) and multi-turn factual audits (TD-EVAL; Acikgoz et al., 2025) inspired pairing de-lexicalized SBERT style cosine with factual Slot F_1 and Exemplar Leakage Rate; a priori noncentral-*F* power derivations (Cohen, 1988; Faul et al., 2007) and Greenhouse–Geisser sphericity corrections (Greenhouse & Geisser, 1959) governed sample allocation across all four stages; and heteroscedasticity-consistent (HC3) and non-parametric Aligned Rank Transform (ART) factorial models in *Statsmodels* (Seabold & Perktold, 2010; Wobbrock et al., 2011) verified every 2 × 2 (**RQ3-Govern**) and 2 × 3 (**RQ4-Optimize**) interaction under boundary floors. Operationalizing these four literature streams into a unified pipeline directly bridges four sequential research gaps: (1) prefix-truncated LLM-hybrid archetype discovery and early routing (**RQ1-Identify stage**) → (2) archetype-partitioned FAISS style and pacing personalization (**RQ2-Personalize stage**) → (3) 2 × 2 neuro-symbolic YAML rule governance and slot masking (**RQ3-Govern stage**) → (4) 2 × 3

multi-turn turn-economy optimization via informed-slot ledgers and persona trade-offs (**RQ4-Optimize stage**).

Materials and Methods

Data Sources, Acquisition, and Inclusion Criteria

This study utilizes two publicly available, peer-reviewed English task-oriented dialogue corpora (Table 3): Microsoft Frames (El Asri et al., 2017; 1,369 raw dialogues collected in August–September 2016 and published in 2017; 1,334 analytic dialogues) and MultiWOZ 2.2 (Budzianowski et al., 2018; Zang et al., 2020; 10,438 raw dialogues published in 2020, from which a seeded travel-domain sample of 1,500 dialogues yielded 1,499 analytic dialogues). No proprietary web scraping was required; raw JavaScript Object Notation (JSON) archives were acquired programmatically via *src/ingestion.py* from the official Microsoft Research Open Data / Hugging Face repository (https://huggingface.co/datasets/google/frames-benchmark) and the official MultiWOZ 2.2 GitHub repository (https://github.com/budzianowski/multiwoz). Both datasets were collected under institutional research consent, assign anonymized alphanumeric role-player identifiers, and contain zero personally identifiable information.

Explicit **inclusion and exclusion criteria** were enforced programmatically during ingestion (*src/final_pipeline/step1_prepare_data.py*; Table 5). Dialogues were *included* if they (a) contained valid dialogue and speaker identifiers, (b) involved vacation package negotiation (Frames) or travel/local hospitality services (hotel, train, attraction, or taxi in MultiWOZ 2.2), and (c) comprised at least three alternating turns (*total_turns* ≥ 3, ensuring at least two user turns and one assistant turn for prefix routing). Dialogues were *excluded* if they (a) were exact duplicate transcripts detected via SHA-256 text hashing (1 excluded in Frames; 0 in MultiWOZ) or (b) were aborted fragments with fewer than three turns (34 excluded in Frames, leaving 1,334

analytic dialogues; 1 excluded in MultiWOZ, leaving 1,499 analytic dialogues). Table 4 defines all analytical variables and measurement units (raw corpus fields and canonical schema mappings are documented in Appendix A, Table A3).

Table 3

Dataset Overview, Public Acquisition URLs, Timeframes, and Analytical Roles

Attribute	Microsoft Frames (primary corpus)	MultiWOZ 2.2 (comparison corpus)
Citation & timeframe	El Asri et al. (2017); dialogues recorded Aug–Sep 2016, published 2017; analyzed 2026	Budzianowski et al. (2018); Zang et al. (2020); corrected release published 2020; analyzed 2026
Public acquisition URL & format	https://huggingface.co/datasets/google/frames-benchmark (single *frames.json* archive)	https://github.com/budzianowski/multiwoz (sharded *dialogues_*.json* archives)
Sample size & unit of analysis	1,369 raw; 1,334 analytic multi-turn dialogue trajectories (golden D_{gold} = 933, held-out D_{test} = 401)	10,438 raw; seeded travel sample of 1,500; 1,499 analytic multi-turn dialogue trajectories
Domain & task setting	Consultative flight + hotel vacation package negotiation under budget, date, and star-rating constraints	Multi-domain city services (hotel, train, taxi, attraction) slot-filling without flight packages
Annotations extracted	Utterance text, speaker role, dialogue acts, slot–value arguments, wizard task success, user 1–5 rating	Utterance text, speaker role, active service domains, turn-level belief-state slot dictionaries
Analytical role in study	RQ1 clustering, feature importance, & prefix router; RQ2 prompting; RQ3 2×2 governance; RQ4 2×3 simulation	External structural comparison in EDA (Tables 5–6, Figure 1) and cross-corpus projection test in RQ1

Table 4

Data Dictionary of Analytical Variables With Units and Operational Definitions

Variable	Role	Type and unit	Operational definition and scale	RQ
Clustering algorithm	Primary IV	Categorical (3 levels)	*K*-Means (*k*=3), Ward agglomerative (*k*=3), HDBSCAN	1
Silhouette & Davies–Bouldin	Primary/Sec. DV	Continuous (unitless / ratio)	Cosine Silhouette [−1, 1] and cluster scatter ratio (≥ 0)	1
Adjusted Rand index (ARI)	Diagnostic DV	Continuous [−1, 1] (unitless)	Chance-corrected partition stability across 80% subsamples	1
LLM behavioral profile (16 fields)	Clustering / Audit	10 ordinal (1–5) + 6 cat./bool.	10 Likert scales + 3 goal/tone clustering fields + 3 audit fields (Table C1)	1, 4
Feature importance (η^2, $\|\beta\|_2$, $\Delta\kappa$)	Explanatory metric	Proportion [0, 1] / L2 norm / $\Delta\kappa$	Cluster variance explained η^2, router weight norm, & test permutation κ drop	1, 2

Variable	Role	Type and unit	Operational definition and scale	RQ
Refined user archetype	IV / Moderator	Categorical (3 levels)	*Decisive Booker*, *Open Explorer*, *Rigid-Constraint Seeker*	1–4
Prefix routing κ & ROC-AUC	Validation DV	Continuous [−1, 1] / [0, 1]	Chance-corrected Cohen's κ and macro One-vs-Rest area under ROC curve	1
Prompting strategy	Primary IV	Categorical (2 or 3 levels)	Zero-Shot, Static Few-Shot (k=3), Cluster-Adaptive Few-Shot (k=3)	2–4
Style convergence & Pacing err.	Primary/Sec. DV	Cosine [−1, 1] / Words (count)	De-lexicalized SBERT cosine to archetype centroid; $\lvert words_{gen} - \mu_{words,a} \rvert$	2, 3
Rule-engine mode	Primary IV	Binary (2 levels)	*Rules Off* (unguarded) vs. *Rules Active* (YAML check + [SLOT] mask + guard)	3
Rule adherence rate (RAR)	Primary DV	Proportion [0, 1] (rate)	Violation enforcement sensitivity (n=18/cell), specificity (n=18/cell), & overall	3
Exemplar leakage rate (ELR)	Primary DV	Proportion [0, 1] (rate)	Share of response entities originating strictly from prompt exemplars	3
Factual slot coverage F_1	Primary DV	Proportion [0, 1] (F_1 score)	Harmonic mean of precision and recall over active booking state slots	3
Turns to goal (ln TGC) & RER	Primary/Sec. DV	Log count (ln turns) / Rate [0, 1]	Natural log of total turns until [ACCEPT_BOOKING]; redundant asks / agent turns	4

Note. IV = independent variable; DV = dependent variable; ROC-AUC = receiver operating characteristic area under curve. Raw and canonical schema fields are in Appendix A, Table A3.

Table 5

Data Cleaning, Inclusion/Exclusion Criteria, and Quality Audit Log

Audit check or criterion	Target fields	Detection rule	Frames	MultiWOZ	Treatment and analytical rationale
Exact duplicate transcripts	Utterance text	SHA-256 full-text hash	0	1	Excluded duplicate record to prevent double-weighting in clustering
Minimum turn floor (< 3 turns)	total_turns	Turn count < 3	35	0	Excluded aborted fragments; ≥ 3 turns required for prefix routing
Upper-tail turn outliers	total_turns	$> Q_3 + 1.5 \times$ IQR	22 (> 32.5)	13 (> 27.0)	Retained as authentic complex negotiations; natural-log transformed in RQ4

Audit check or criterion	Target fields	Detection rule	Frames	MultiWOZ	Treatment and analytical rationale
Unobserved optional slots	Tier 1 & Tier 2 slots	Slot never stated in chat	Table E1	Table E1	Preserved semantically as None; never mean/mode-imputed
Missing post-chat user rating	user_rating	Null survey field	3 (11 raw)	N/A	Retained for unsupervised clustering; excluded only from rating checks
Surface entity values	Utterance text	Slot & numeric regex	All	All	Replaced with [SLOT] tags in FAISS exemplars to prevent entity leakage
Final analytic sample	All schema fields	Post-cleaning count	1,334 (933/401)	1,499	Locked with stratified 70/30 golden/test split (seed = 42)

Note. IQR = interquartile range. Zero duplicate dialogue IDs or empty utterances occurred in either cleaned corpus.

Database Setup, Vector Indexing, and Data Access

To support reproducible offline experimentation and sub-millisecond online retrieval, the data layer is organized into a four-component storage architecture accessed via typed Python loaders (*src/ingestion.py*, *src/final_pipeline/common.py*, and *src/final_pipeline/routing.py*):

1. **Canonical Document Database (JSON Lines [JSONL] & Pydantic):** Raw JSON corpora are normalized into a two-tier *Pydantic* 2.9 schema (*src/canonical_schema.py*) and persisted as line-delimited JSON (*data/canonical/canonical_corpus.jsonl*) alongside indexed tabular views (*data/interim/frames_clean.csv* and *multiwoz_clean.csv*). Tier 1 stores nine core travel slots (origin city, destination city, departure date, return date, duration in nights, adult count, child count, budget in United States dollars [USD], and accommodation star tier), and Tier 2 stores five enterprise policy attributes (visa lead window in days, passport expiry in months, package class [Free Independent Traveler (FIT) vs. Group], room-sharing basis, and promotional discount percentage).

2. **Partitioned Vector Database (FAISS IndexFlatIP):** Dense embedding matrices are stored as memory-mapped NumPy arrays (*data/interim/*.npy* and content-addressed SHA-1 caches in *data/embeddings/gemini/cache/*). At runtime, *RefinedArchetypeRouter* indexes L2-normalized 46-dimensional *Blocks* vectors from successful golden-partition dialogues into one global baseline index and three archetype-partitioned *faiss.IndexFlatIP* inner-product sub-indices (*Decisive Booker*, *Open Explorer*, and *Rigid-Constraint Seeker*), executing exact cosine nearest-neighbor exemplar lookup in under 2 ms.
3. **Declarative Policy Store & Session State Ledger:** Enterprise business rules are externalized in a version-controlled YAML database (*rules/rules_enterprise.yaml*) evaluated deterministically by *DeclarativeRuleEngine* (*src/rule_engine.py*). Active multi-turn sessions maintain a structured *Verified Informed Slot Ledger* in memory (or Redis in production) paired with content-addressed JSONL response caches (*data/final/*.jsonl*) that guarantee 100% deterministic replay.

Data splitting procedure. To prevent any information leakage between archetype discovery and downstream evaluation, the 1,334 analytic Frames dialogues were partitioned once prior to modeling (random seed = 42, stratified on binary wizard task completion) into a **70% golden discovery and training partition ($\boldsymbol{D_{\text{gold}}}$, $\boldsymbol{n = 933}$)** and a **30% held-out test partition ($\boldsymbol{D_{\text{test}}}$, $\boldsymbol{n = 401}$)**. Every unsupervised and supervised parameter, including *StandardScaler* means and variances, 20-component Gemini principal component analysis (PCA) projection matrices, *K*-Means cluster centroids, L2 multinomial logistic regression router weights, FAISS exemplar pools, and target archetype word-length norms, was fitted strictly on D_{gold}. The held-out partition D_{test} was accessed exclusively for out-of-sample prefix routing evaluation (RQ1) and downstream prompting and simulation experiments (RQ2–RQ4).

Exploratory Data Analysis

Univariate and Outcome-Stratified Analysis

Table 6 and Figure 1 summarize the univariate distributions across corpora, partitions, commercial trip attributes, and session outcomes. Microsoft Frames and MultiWOZ 2.2 exhibit comparable mean conversation lengths (M = 14.93 vs. 15.17 turns; Panel A), but Frames displays substantially wider standard deviation dispersion (SD = 6.74 vs. 4.46, range 3–48 vs. 4–32 turns) and higher assistant verbosity (M = 15.0 vs. 15.1 words/turn; Panel B), reflecting complex multi-package comparisons. Covariate balance checks between D_{gold} (n = 933) and D_{test} (n = 401) confirm zero statistically significant differences across dialogue turns (t = 0.13, p = .899), slot revisions (t = 1.04, p = .299), package offers (t = 0.96, p = .338), user words per turn (p = .286), or behavioral ratings ($p \geq .475$). Across stated numeric trip budgets in Frames (n = 693; Figure 1, Panel C), budgets span $200 to $54,700 USD with a right-skewed mean of **$5,338** ($SD$ = $6,402) and median of **$3,400** (interquartile range [IQR] = $2,800). Traveler parties average 11.33 adults (SD = 337.79; solo = 28.4%, group = 18.7%, family = 17.5%, couple = 9.7%), and post-chat user satisfaction averages 4.58/5.0 (SD = 0.83, n = 1331). Canonical slot completeness (Appendix E, Table E1 and Figure E1) exceeds 70%–99% for core Tier-1 slots in Frames (origin 99.3%, destination 99.0%, departure date 85.0%, budget 73.1%, hotel tier 72.9%) but is **0.0%** for Tier-2 visa, passport, and discount rules in both archival corpora, directly motivating controlled Tier-2 state injection in RQ3.

Crucially, stratifying the Frames corpus by session outcome (Table 6, bottom rows; Figure 1, Panel D) reveals two distinct negotiation failure modes across *Booked Package* (65.6%, n = 875), *No Suitable Option* (28.1%, n = 375), and *Undecided / Left* (6.2%, n = 83). One-way ANOVA and Kruskal–Wallis tests confirm highly significant differences across outcomes in

total turns ($F(2, 1330) = 175.39$, $p < .001$), slot revisions ($F = 110.80$, $p < .001$), package offers ($F = 145.98$, $p < .001$), destination flexibility ($F = 37.00$, $p < .001$), exploration breadth ($F = 65.41$, $p < .001$), and decisiveness ($F = 138.50$, $p < .001$). Specifically, *No Suitable Option* sessions involve rigid, low-flexibility shoppers (destination flexibility = 2.18 vs. 2.77 for booked) who receive only **1.03 package offers** before hitting an impasse at 10.23 turns, whereas *Undecided / Left* sessions involve high-exploration, low-decisiveness shoppers (exploration = 2.69, decisiveness = 2.35 vs. 3.71 for booked) who churn through 17.00 revisions and 3.87 offers without closing, directly motivating persona-specific trade-off and comparison policies.

Table 6

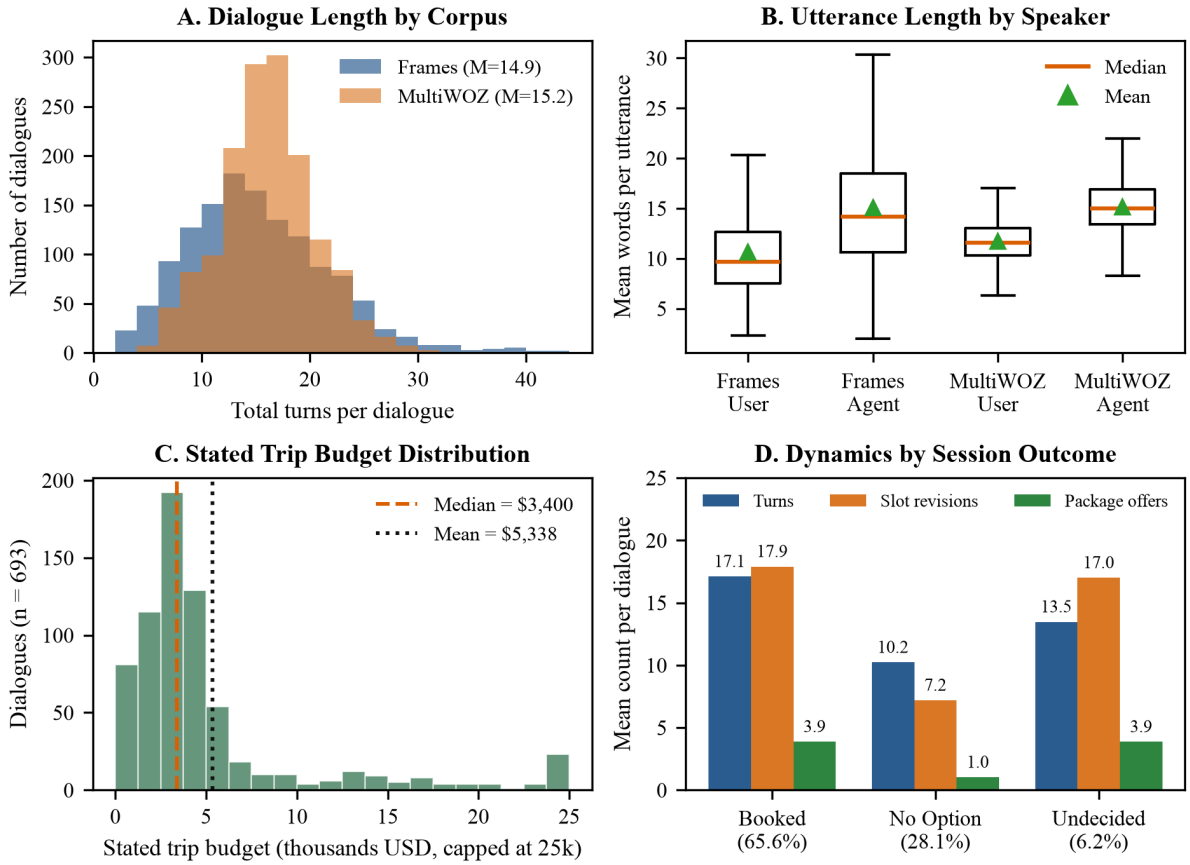
Univariate and Outcome-Stratified Descriptive Statistics of the Analytic Samples

Sample or outcome stratum	*N* (%)	Turns *M* (*SD*)	*Mdn* / Rev.	Range / Offers	User wds / Flex.	Agent wds / Dec.	Success / Budget
Microsoft Frames	1,334	14.93 (6.74)	14	3–48	10.6	15.0	94.2%
MultiWOZ 2.2	1,499	15.17 (4.46)	16	4–36	11.7	15.1	N/A
Frames golden partition (D_gold)	933	14.95 (6.75)	14	3–48	10.7	15.3	94.2%
Frames held-out partition (D_test)	401	14.90 (6.71)	14	3–43	10.4	14.5	94.3%
Frames Outcome: Booked Package	875 (65.6%)	17.10 (6.26)	Rev: 17.9	Off: 3.87	Flex: 2.77	Dec: 3.71	$3,700
Frames Outcome: No Suitable Option	375 (28.1%)	10.23 (5.30)	Rev: 7.2	Off: 1.03	Flex: 2.18	Dec: 2.91	$3,100
Frames Outcome: Undecided / Left	83 (6.2%)	13.47 (5.98)	Rev: 17.0	Off: 3.87	Flex: 3.19	Dec: 2.35	$3,800

Note. Top four rows report corpus/partition distributions (user/agent words per turn and wizard task success %). Bottom three rows report Frames outcome strata ($N = 1,333$ excluding 1 unclear session): Rev. = mean slot revisions; Offers = mean package offers; Flex. = destination flexibility (1–5); Dec. = decisiveness (1–5); Budget = median stated budget (USD).

Figure 1

Univariate Dialogue Length, Utterance Length, Stated Trip Budget, and Session Outcome Dynamics



Note. Frames $N = 1{,}334$ (stated budgets $n = 693$); MultiWOZ 2.2 $N = 1{,}499$.

Bivariate Analysis

Parametric (Pearson r) and non-parametric (Spearman ρ) bivariate correlations across all 1,334 Frames dialogues (Table 7; Figure 2) demonstrate that conversation length is heavily driven by cumulative slot revisions ($r = .74$, $\rho = .76$, $p < .001$; Panel A) and assistant package offers ($r = .56$, $\rho = .61$, $p < .001$). Cross-checking structural dialogue counts against the opening LLM behavioral ratings establishes strong construct validity: slot revisions correlate positively with exploration breadth ($r = .60$, $\rho = .67$, $p < .001$), user words per turn correlate positively with

upfront information provision ($r = .16$, $\rho = .21$, $p < .001$), and user decisiveness correlates negatively with dialogue length ($r = -.31$, $\rho = -.29$, $p < .001$) and positively with package booking completion ($r = .40$, $\rho = .40$, $p < .001$). Across archetypes (Figure 2, Panel B), *Open Explorers* average 24.9 slot revisions and 5.0 package offers per session, whereas *Rigid-Constraint Seekers* average only 7.7 revisions and 1.3 offers.

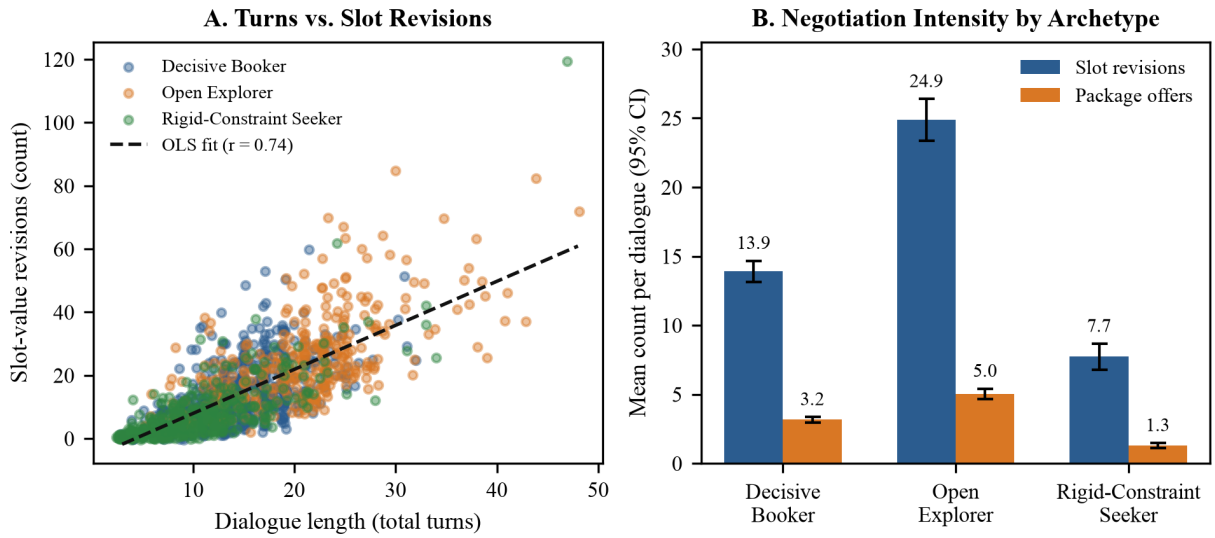
Table 7

Bivariate Parametric and Non-Parametric Correlations of Dialogue Dynamics (Frames, N = 1,334)

Variable pair	Pearson *r*	*p* (Pearson)	Spearman ρ	*p* (Spearman)
Total Turns vs. Slot Revisions	.741	< .001	.759	< .001
Total Turns vs. Agent Package Offers	.561	< .001	.608	< .001
Total Turns vs. User Hesitation Markers	.336	< .001	.355	< .001
Slot Revisions vs. Exploration Breadth Rating	.603	< .001	.673	< .001
Total Turns vs. User Decisiveness Rating	−.310	< .001	−.289	< .001
User Words/Turn vs. Information Up Front Rating	.158	< .001	.206	< .001
Agent Offers vs. Destination Flexibility Rating	.324	< .001	.358	< .001
Question-Asking Rating vs. Total Turns	.558	< .001	.603	< .001
User Decisiveness Rating vs. Package Booking Indicator	.398	< .001	.399	< .001

Figure 2

Bivariate Relationships Between Dialogue Length, Slot Revisions, and Archetype Negotiation Intensity



Note. Panel A: Dialogue turns vs. slot revisions with OLS regression trend. Panel B: Mean revisions and package offers by refined archetype (error bars = 95% confidence interval [CI]).

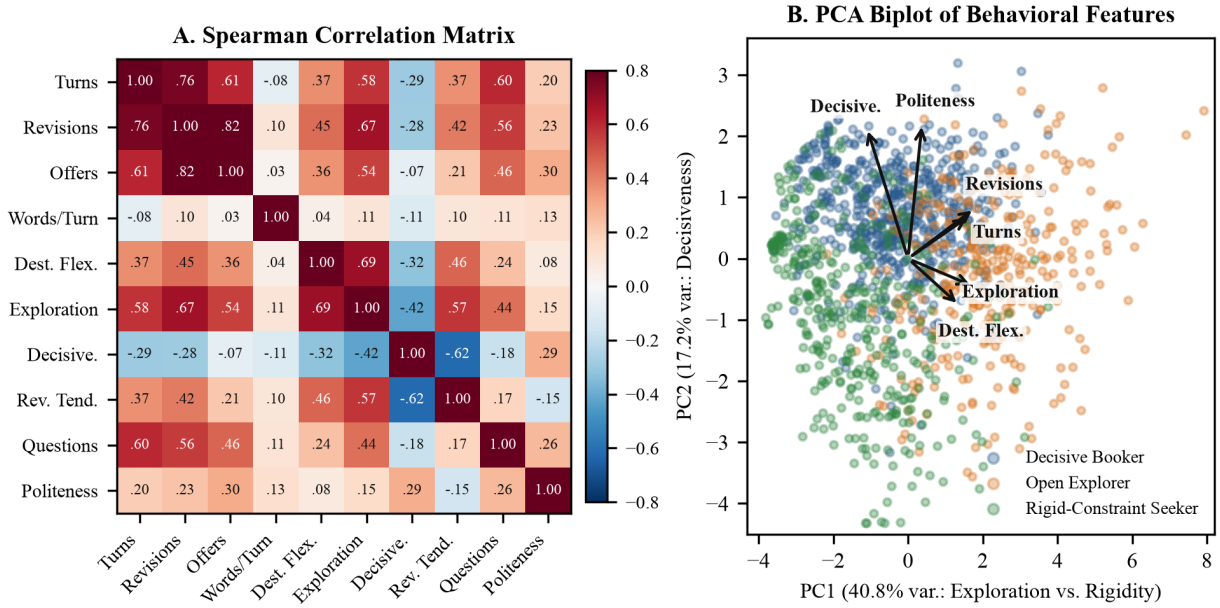
Multivariate Analysis

Multivariate structure across ten structural and behavioral variables was analyzed via Spearman rank correlation and principal component analysis (PCA; Figure 3; full component loadings in Appendix E, Table E2). The first three principal components account for 68.6% of total variance: PC1 (40.8%) represents an *Exploration vs. Constraint Rigidity* axis loaded on slot revisions (.42), total turns (.39), exploration breadth (.39), and question asking (.34); PC2 (17.2%) represents a *Decisiveness and Politeness* axis (−.51, −.46); and PC3 (10.6%) isolates *User Utterance Verbosity* (.89). Projecting dialogues onto the (PC1, PC2) plane (Figure 3, Panel B) confirms that the three refined archetypes occupy distinct regions of a continuous behavioral manifold rather than isolated clusters. ***EDA → RQ1-Identify & RQ3-Govern stage handoff (dropped vs. carried forward):*** Because whole-dialogue turn counts, cumulative revisions, and

end-of-session booking outcomes are endogenous consequences of negotiation failure rather than observable at session start, they are **dropped** as clustering inputs to prevent circular outcome confounding; instead, opening-prefix utterance representations ($M \leq 9$ user messages) and non-outcome behavioral scales are **carried forward into the RQ1-Identify stage**, while controlled Tier-2 policy state injection is **carried forward into the RQ3-Govern stage** to overcome the 0.0% archival Tier-2 slot coverage.

Figure 3

Multivariate Spearman Correlation Matrix and Principal Component Biplot of Behavioral Features



Note. Panel A: 10 × 10 Spearman rank correlation matrix. Panel B: PCA biplot colored by refined archetype with top loading vectors.

Research Design, Hypotheses, and Sample Size Justification

All four research questions are evaluated at a two-tailed significance level of **α = .05** against the formal null (H_{01}–H_{04}) and alternative (H_{a1}–H_{a4}) hypotheses specified in Table 1. Minimum sample sizes were derived a priori using Cohen's medium effect size $f = .25$ (Cohen,

1988), $\alpha = .05$, and target statistical power $1 - \beta \geq .80$ via the noncentral F distribution (Faul et al., 2007):

$$1 - \beta = 1 - F_{nc}(F_{crit}; df_1, df_2, \lambda), \quad F_{crit} = F_{1-\alpha}(df_1, df_2) \quad (1)$$

where $\lambda = f^2N$ for between-subjects designs (RQ1, RQ3, RQ4) and $\lambda = f^2Nm\varepsilon / (1 - \rho)$ for repeated-measures RQ2 ($m = 3$ conditions, $\rho = .50$, $\varepsilon = 1.0$). Table 8 reports the exact degrees of freedom, noncentrality parameter λ, critical value F_{crit}, minimum sample size N_{min}, and buffered balanced sample size N_{final} (achieving actual power of .805–.878). Crucially, these computed sample sizes directly justify how the collected corpus volume ($N = 1{,}334$ analytic Frames dialogues) is allocated across the four stages: (1) in the **RQ1-Identify** stage, $N_{min} = 159$ evaluations across 3 algorithms is satisfied by drawing 53 matched subsamples of $n = 500$ dialogues from D_{gold} ($n = 933$, a 53.6% finite-population fraction paired with matched repeated-measures ANOVA and Friedman tests across the 53 shared seeds); (2) in the **RQ2-Personalize** stage, $N_{min} = 28$ repeated-measures subjects is exceeded by sampling $N_{final} = 33$ stratified held-out dialogues from D_{test} (11 per archetype × 3 conditions = 99 generations); (3) in the **RQ3-Govern** stage, $N_{min} = 128$ evaluations for a 1-*df* numerator effect in a 2 × 2 design is exceeded by evaluating $N_{final} = 144$ units (36 stratified held-out dialogues, comprising 18 policy-violating and 18 compliant cases at 12 per archetype, tested across all four cells, $n = 36$/cell); and (4) in the **RQ4-Optimize** stage, $N_{min} = 158$ episodes for a 2-*df* prompting × archetype interaction in a 2 × 3 design is exceeded by allocating $\boldsymbol{N_{final} = 180}$ **distinct, non-overlapping held-out seed dialogues** from D_{test} ($n = 401$; 60 per archetype, partitioned into $n = 30$ unique dialogues for *Zero-Shot* and $n = 30$ unique dialogues for *Cluster-Adaptive Few-Shot*), ensuring strict between-subjects independence ($df_{error} = 174$).

Table 8

Sample Size and Noncentral-F Power Calculations Matched to Each Planned Analysis

RQ	Planned experimental design	*df* at N_{min}	λ	F_{crit}	N_{min}	Power	N_{final} (per cell)	Final power
RQ1	One-way ANOVA (3 algorithms)	2, 155	9.88	3.05	158	.802	159 (53)	.805
RQ2	One-way repeated-measures ANOVA (3 conditions)	2, 54	10.50	3.17	28	.812	33 (11)	.878
RQ3	2 × 2 factorial ANOVA	1, 124	8.00	3.92	128	.801	144 (36)	.846
RQ4	2 × 3 factorial ANOVA	2, 152	9.88	3.06	158	.802	180 (30)	.855

Note. Computed for medium effect size $f = .25$ at $\alpha = .05$ and target power $1 - \beta \geq .80$ (*src/final_pipeline/step0_sample_size.py*).

Model Selection, Feature Importance Assessment, and Evaluation Metrics

Model selection justification and hybrid representation (RQ1-Identify stage). Three clustering families representing distinct geometric assumptions were selected for benchmarking: centroid-based *K*-Means (spherical Voronoi cells guaranteeing 100% point assignment), hierarchical Ward agglomerative clustering (bottom-up variance minimization), and density-based HDBSCAN (arbitrary-shape density cores with noise rejection). Primary whole-dialogue vectors pool de-lexicalized turn embeddings $\mathbf{e}_t$ with exponential recency decay ($\lambda_d = 0.15$):

$$\mathbf{v} = \Sigma_{t=1}^{T} w_t \mathbf{e}_t, \quad w_t = \exp[-\lambda_d(T - t)] \,/\, \Sigma_{j=1}^{T} \exp[-\lambda_d(T - j)] \quad (2)$$

To eliminate end-of-dialogue outcome confounding, the refined representation (*Blocks*) concatenates two variance-equalized blocks computed strictly on the first *M* user messages: (1) the top 20 principal components (*PCA-20*, fitted on D_{gold}) of the 3,072-dimensional Gemini embedding (*gemini-embedding-001*; Lee et al., 2025), and (2) a standardized 26-column LLM behavioral matrix encoding 13 clustering fields from the 16-field *UserBehaviour* schema (ten 1–5 ordinal scales, six primary-goal indicators, six secondary-goal indicators, and four emotional-

tone indicators; three outcome/party audit fields are excluded from clustering; Appendix C, Tables C1–C2; test–retest weighted $\kappa = .87–.96$; revision criterion correlation $\rho = .73$).

Feature importance assessment and analytical utilization. To state explicitly how feature importance is assessed and used across the pipeline, every dimension $j \in \{1, ..., 46\}$ of the golden-fitted *Blocks* representation is evaluated using three complementary importance metrics (*src/final_pipeline/step5_figures_and_eda.py*; results in Table 13 and Figure 6):

1. **Unsupervised Cluster-Discriminative Importance (η^2 and ANOVA F on D_{gold}):** For each feature j, one-way ANOVA across the three K-Means reference clusters on D_{gold} ($n = 933$) computes the proportion of total variance explained by cluster membership, $\eta_j^2 = SS_{between,j} / SS_{total,j}$.
2. **Supervised Router Coefficient Norm ($\|\boldsymbol{\beta}_j\|_2$ on D_{gold}):** In the L2-regularized multinomial logistic regression router ($C = 1.0$) fitted on standardized golden features, the Euclidean norm across the three class weight vectors, $\|\boldsymbol{\beta}_j\|_2 = (\Sigma_{c=1}^{3} \beta_{c,j}^2)^{1/2}$, quantifies feature j's direct leverage on the classification decision boundary at prefix horizons $M = 5$ and $M = 9$.
3. **Out-of-Sample Permutation Importance ($\Delta\kappa$ on Held-Out D_{test}):** To measure predictive generalization without training bias, each feature (and each of the four feature blocks: *20 Gemini Semantic PCs*, *10 Ordinal Behavioral Scales*, *12 Primary/Secondary Goals*, and *4 Emotional Tone Indicators*) is randomly permuted 50 times on held-out D_{test} ($n = 401$), recording the mean drop in chance-corrected routing agreement $\Delta\kappa = \kappa_{baseline} - \kappa_{permuted}$.

These feature importance results are used directly in four analytical decisions: (a) *representation pruning and block weighting (RQ1-Identify stage)*, confirming that both the

Gemini semantic block ($\Delta\kappa = .310$ at $M = 5$; .674 at $M = 9$) and the LLM behavioral block ($\Delta\kappa = .215$ at $M = 5$; .394 at $M = 9$) are essential while excluding the three outcome/party audit fields prevents target leakage; (b) *deterministic cluster naming (RQ1-Identify stage, Appendix F)*, where the top orthogonal behavioral features (*exploration_breadth* / *destination_flexibility*, $\eta^2 = .304/.150$, and *decisiveness*, $\eta^2 = .280$) define the programmatic centroid-labeling rules in Table F1; (c) *prefix router calibration (RQ1-Identify → RQ2-Personalize stages)*, where separated training (D_{gold}), 5-fold cross-validation, and held-out test (D_{test}) evaluations verify that high-importance features generalize without overfitting; and (d) *persona policy and simulator design (RQ2-Personalize & RQ4-Optimize stages)*, where the highest-η^2 features per cluster parameterize the assistant's pacing directives and the RQ4 blind persona validation rubric. Tables 9 and 10 define the five declarative YAML policy rules and all evaluation metrics.

Table 9

Declarative Enterprise Policy Rules Evaluated in RQ3

Rule ID and name	Trigger condition	Required action	Severity
RULE_VISA_WINDOW: Visa Processing Lead Time	departure_lead_days < 15; is_international = true	block and flag	fatal
RULE_PASSPORT_VALIDITY: Passport Validity Threshold	passport_expiry_months < 6; is_international = true	block and flag	fatal
RULE_PAX_CAPACITY: Maximum Adult Capacity per Single Booking	traveler_count_adults > 8; package_type = 'FIT'	convert to group	warning
RULE_DISCOUNT_THRESHOLD: Maximum Discretionary Sales Discount	discount_applied_percentage > 15.0	clamp discount	fatal
RULE_BUDGET_MINIMUM: Minimum Feasible Per-Pax International Budget	budget stated; budget_amount < 500; is_international = true	prompt realistic budget	warning

Table 10

Evaluation Metrics, Mathematical Formulae, and Analytical Interpretation

Metric	Mathematical formula	RQ	Analytical interpretation
Silhouette & DB index	$s(i) = [b(i)-a(i)]/\max\{a,b\}$; $DB = (1/k)\Sigma \max(\sigma_i+\sigma_j)/d(\mathbf{c}_i,\mathbf{c}_j)$	1	Higher Silhouette [−1, 1] and lower DB (≥ 0) indicate superior separation
ARI, Cohen's κ, & ROC-AUC	$ARI = (RI-\mathrm{E}[RI])/(\max RI-\mathrm{E}[RI])$; $\kappa = (p_o-p_e)/(1-p_e)$; OvR AUC	1, 3, 4	Bootstrap stability (ARI ≥ .60), chance-corrected routing/judge κ, & class discrimination
Feature importance	$\eta_j^2 = SS_{B,j}/SS_{T,j}$; $\|\boldsymbol{\beta}_j\|_2 = \sqrt{\Sigma\beta_{c,j}^2}$; $\Delta\kappa_j = \kappa - \kappa_{\mathrm{perm},j}$	1, 2	Cluster variance explained, router weight magnitude, & held-out permutation κ drop
Style cosine & Pacing err.	$\cos(\mathbf{r}, \mathbf{c}_a) = (\mathbf{r}\cdot\mathbf{c}_a)/(\|\mathbf{r}\|\|\mathbf{c}_a\|)$; $\lvert W_{\mathrm{gen}} - \mu_{W,a}\rvert$	2, 3	De-lexicalized SBERT structural style cosine and word-count pacing error
Rule adherence (RAR)	Sens. = $\lvert R_{\mathrm{enf}} \cap R_{\mathrm{viol}}\rvert/\lvert R_{\mathrm{viol}}\rvert$ (n=18/cell); Spec. = 1 − false_block (n=18/cell)	3	Decomposed into Violation Enforcement Sensitivity, Compliant Specificity, & Overall
Exemplar leakage (ELR)	$\lvert E_{\mathrm{resp}} \cap (E_{\mathrm{ex}} \setminus E_{\mathrm{diag}})\rvert$ / $\max(1, \lvert E_{\mathrm{resp}}\rvert)$	3	Proportion [0, 1] of response entities leaked from prompt exemplars
Slot coverage F_1	2 · Precision · Recall / (Precision + Recall) over active state slots	3	Proportion [0, 1] of active booking slots grounded in generated response
Turns to goal & RER	TGC = total turns (user + assistant) to [ACCEPT_BOOKING] (min = 5); RER = A_{red} / A_{tot}	4	Lower total/log conversational turns and lower RER indicate superior turn economy

Note. Parametric ANOVAs are paired with Welch/HC3 robust tests and non-parametric (Kruskal–Wallis, Friedman, Mann–Whitney, ART; Wobbrock et al., 2011) checks.

Architecture and Workflow

System Overview

The proposed neuro-symbolic conversational system is structured as an end-to-end seven-stage pipeline that strictly separates offline discovery and training on the golden partition (D_{gold},

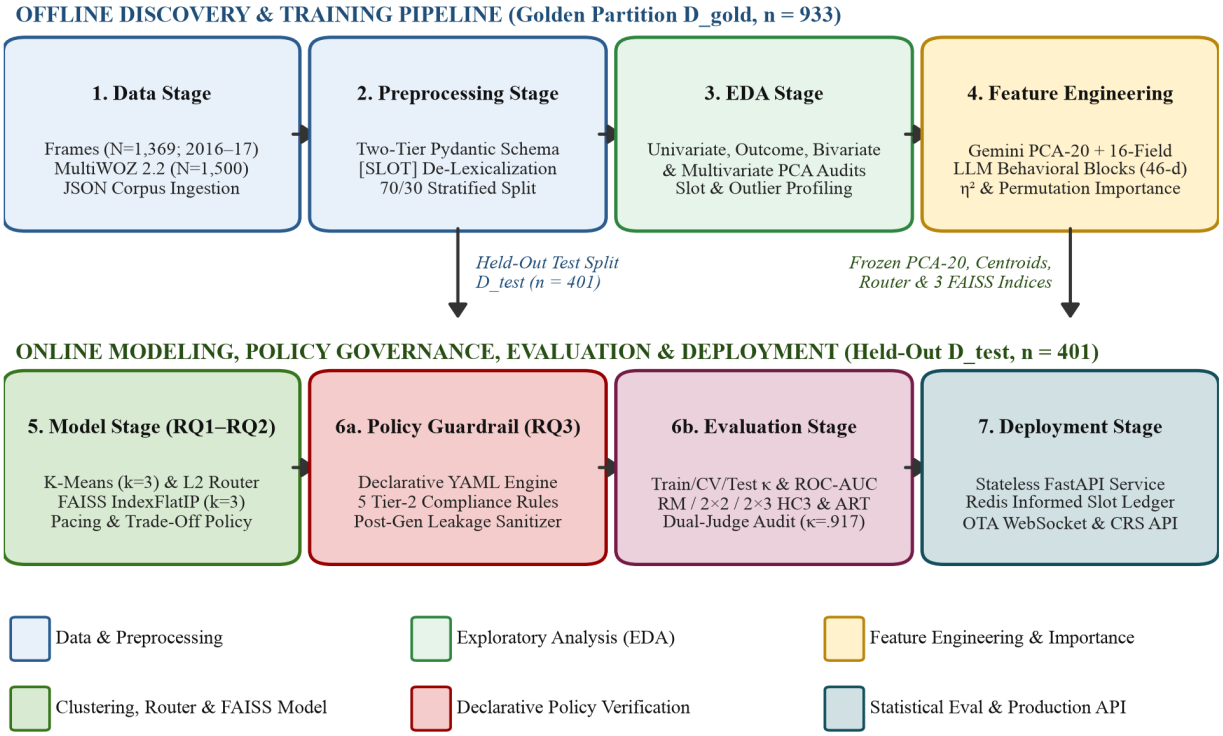
n = 933) from online adaptive inference, policy governance, and evaluation on the held-out test partition (D_{test}, n = 401). The purpose of the pipeline is to resolve the core trade-off identified in the research problem: adapting assistant style and recommendation pacing to heterogeneous traveler archetypes early in a conversation while guaranteeing zero exemplar entity leakage, deterministic booking-policy compliance, and minimal turns to quotation agreement.

Architecture Diagram

As shown in Figure 4, the system architecture progresses through seven explicit stages: (1) *Data Stage*, (2) *Preprocessing Stage*, (3) *Exploratory Data Analysis (EDA) Stage*, (4) *Feature Engineering Stage*, (5) *Model Stage*, (6a/6b) *Policy Guardrail and Evaluation Stage*, and (7) *Deployment Stage*, with color-coded functional legends distinguishing data processing, EDA, feature engineering, modeling, symbolic verification, and production serving. Figure 5 details the corresponding four-step experimental execution workflows for RQ1 (Panel A), RQ2 (Panel B), RQ3 (Panel C), and RQ4 (Panel D).

Figure 4

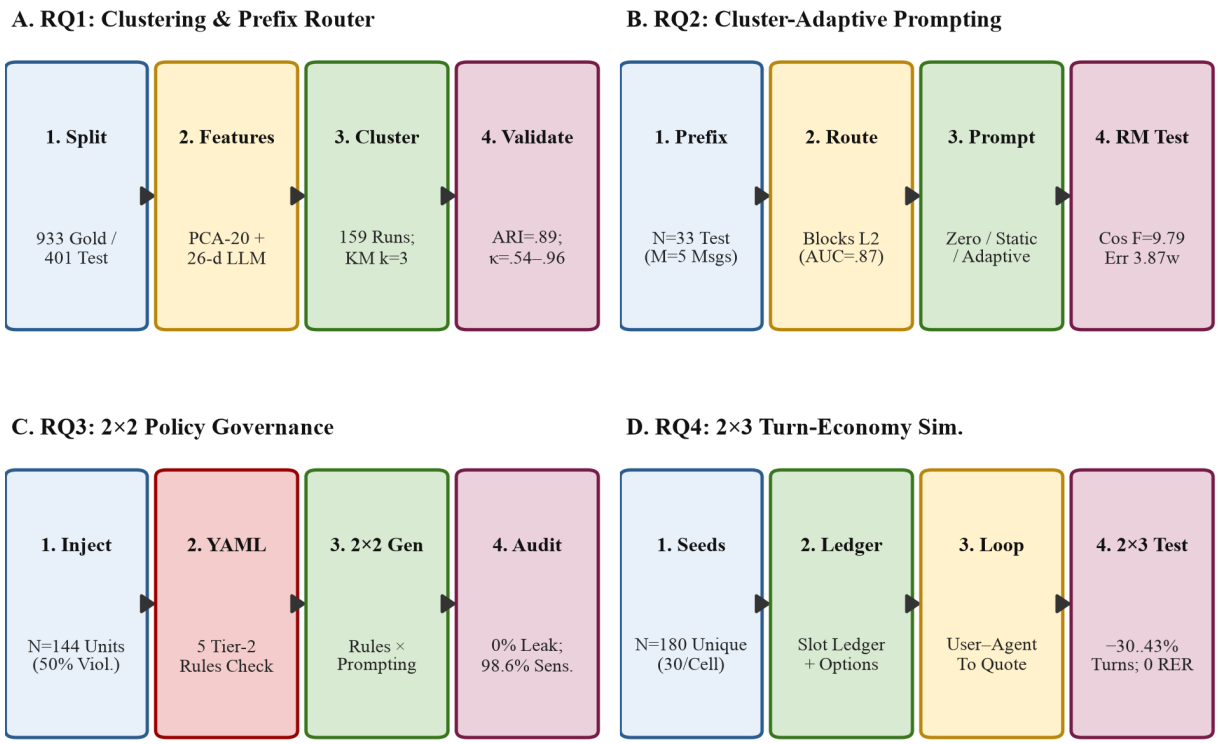
End-to-End Neuro-Symbolic Cluster-Adaptive System Architecture Across All Seven Pipeline Blocks



Note. Pipeline blocks (1)–(4) execute offline on the golden discovery partition (D_{gold}, $n = 933$); pipeline blocks (5)–(7) execute online on held-out test dialogues (D_{test}, $n = 401$) and in production deployment.

Figure 5

Algorithmic Execution Workflows Across Research Questions RQ1 to RQ4



Note. Step-by-step experimental pipelines and statistical hypothesis tests for RQ1 (Panel A), RQ2 (Panel B), RQ3 (Panel C), and RQ4 (Panel D).

Workflow Components

Data ingestion. Raw JSON dialogue archives from Microsoft Frames (1,369 dialogues) and MultiWOZ 2.2 (1,500 seeded travel dialogues) are ingested by *src/ingestion.py*, which parses turn-level speaker tags, utterances, dialogue acts, and belief-state slot dictionaries into validated two-tier *Pydantic* objects stored in *data/canonical/canonical_corpus.jsonl*.

Data preprocessing, scaling, and encoding. In *src/final_pipeline/step1_prepare_data.py* and *step2f_rq1_llm_hybrid.py*, utterance texts are de-lexicalized via slot-value regular expressions that replace specific city names, dates, star ratings, and dollar amounts with generic

[SLOT] tokens. Numerical scaling and categorical encoding follow a strict leakage-free protocol fitted on D_{gold}: (a) 384-d Sentence-BERT and 3,072-d Gemini embedding vectors are L2 unit-norm scaled; (b) the 20 Gemini principal components (*PCA-20*) are centered and projected using golden-partition eigenvectors; (c) the ten 1–5 ordinal LLM behavioral scales are standardized to zero mean and unit variance via *StandardScaler* fitted on D_{gold}; (d) categorical behavioral fields are one-hot or multi-hot encoded into 16 binary columns (6 *primary_goal*, 6 *secondary_goals*, and 4 *emotional_tone* indicators); and (e) in the *Blocks* transformer, the 20-d semantic block and 26-d behavioral block are each scaled by the inverse square root of their golden block trace variance so both modalities contribute exactly equal total variance prior to L2 row normalization.

Missing value and outlier handling. Unobserved optional trip and policy slots (such as *traveler_count_children* or Tier-2 visa/passport fields in archival sessions) represent genuine conversational state absence and are preserved as explicit *None* values in the Pydantic schema rather than imputed, preventing fabricated customer constraints. Missing post-session user ratings (3 analytic Frames dialogues) are retained in clustering and excluded only from rating correlations. Degenerate fragments with fewer than 3 turns (34 in Frames; 1 in MultiWOZ) are removed during cleaning. Upper-tail turn-length outliers exceeding $Q_3 + 1.5 \times \text{IQR}$ (> 29.5 turns in Frames, $n = 30$ [2.2%]; > 28.0 turns in MultiWOZ, $n = 8$ [0.5%]) are retained as authentic complex negotiations; to prevent positive skewness from biasing parametric inference, turn counts are natural-log transformed (ln TGC) in RQ4 and paired with rank-based Kruskal–Wallis, Friedman, Mann–Whitney *U*, and ART tests across all research questions.

Exploratory data analysis and feature engineering. Univariate, outcome-stratified, bivariate, and multivariate PCA diagnostics (Tables 6–7, Figures 1–3) characterize negotiation bottlenecks and motivate combining opening user prefix embeddings ($M \leq 9$ user messages) with

the 16-field LLM behavioral codebook (*step2e_llm_user_annotation.py*). Feature importance is then quantified via golden cluster η^2, standardized router coefficient norms $\|\boldsymbol{\beta}_j\|_2$, and held-out permutation $\Delta\kappa$ (Table 13, Figure 6).

Model development, evaluation, and deployment. Offline model development in the **RQ1-Identify** stage fits *K*-Means ($k = 3$), Ward, and HDBSCAN across 159 resampling runs, locks the three 46-d *Blocks* reference centroids and FAISS *IndexFlatIP* sub-indices on D_{gold}, and trains L2-regularized multinomial logistic prefix routers ($C = 1.0$) verified via 5-fold stratified cross-validation (CV) on D_{gold} and out-of-sample testing on D_{test}. Online evaluation executes the **RQ2-Personalize** repeated-measures prompting harness, the **RQ3-Govern** 2 × 2 rule-governance harness (cross-checked by an independent *gemini-2.5-pro* judge), and the **RQ4-Optimize** 2 × 3 closed-loop turn-economy simulator, which packages into a stateless Python/FastAPI and Redis microservice for live OTA chat gateways.

Tools and Technologies

The experimental pipeline and report automation suite are implemented in Python 3.11 on an Apple M-series workstation (8-core central processing unit [CPU], 16 gigabytes [GB] unified memory) paired with Google Cloud tensor and graphics processing unit (TPU/GPU) inference via the official *google-genai* software development kit (SDK). Data manipulation and schema validation use *pandas* 2.2, *numpy* 1.26, and *pydantic* 2.9. Deep learning embeddings and contextual scoring use *sentence-transformers* (*all-MiniLM-L6-v2*), PyTorch 2.4, and *gemini-embedding-001*; generative experiments and independent validation audits use *gemini-2.5-flash* and *gemini-2.5-pro*; and vector indexing uses *faiss-cpu* 1.8. Unsupervised clustering and manifold projection use *scikit-learn* 1.5 (Pedregosa et al., 2011), *hdbscan* (McInnes et al., 2017), and *umap-learn* (McInnes et al., 2018). Parametric OLS, Greenhouse–Geisser RM-ANOVA,

White/MacKinnon HC3 heteroscedasticity-consistent standard errors, and Aligned Rank Transform (ART) factorial models are computed using *scipy.stats* 1.14 and *Statsmodels* 0.14 (Seabold & Perktold, 2010), while publication figures and American Psychological Association Seventh Edition (APA 7) Word and Portable Document Format (PDF) compilation use *matplotlib* 3.9, *python-docx*, and *PyMuPDF*.

Results

Model Performance

Table 11 presents the consolidated model performance comparison across all four experimental stages, reporting features used, validation and data-splitting methods, separated baseline/training and held-out test metrics, omnibus statistical tests, and explicit identification of the best-performing model in each stage. All four study null hypotheses (H_{01}–H_{04}) are rejected at $\boldsymbol{p < .001}$ (well below $\alpha = .05$). To verify out-of-sample generalization of the supervised prefix router without overfitting, Table 12 reports separated **training ($\boldsymbol{D_{gold}}$, $\boldsymbol{n = 933}$)**, **5-fold stratified cross-validation ($\boldsymbol{D_{gold}}$)**, and **held-out test ($\boldsymbol{D_{test}}$, $\boldsymbol{n = 401}$)** accuracy, chance-corrected Cohen's κ, macro F_1, and macro One-vs-Rest receiver operating characteristic area under the curve (ROC-AUC) across prefix cut-offs $M \in \{3, 5, 7, 9\}$.

Table 11

Consolidated Model Performance Comparison Across Discovery/Training and Held-Out Test Splits

Model / Pipeline stage	Features used	Validation & split	Metric 1 (Train / Baseline)	Metric 2 (Held-Out Test / Proposed)	Key observation & best model
RQ1-Identify (Clustering): ★ K-Means (*k*=3, Best) vs. Ward & HDBSCAN	384-d MiniLM, 3,072-d Gemini, & 46-d *Blocks* (*M*=9)	53 matched *n*=500 seeds on D_{gold} (933) + D_{test} (401)	D_{gold} KM Sil = .147 (100% kept) vs. HDB = .211 (39.1% noise); Blocks DB = 2.647, ARI = .89	D_{test} centroid cos = .523 vs. D_{gold} .526 (p = .676); RM F = 83.04, p < .001	**★ Best: K-Means (*k*=3) on 46-d Blocks** (100% routing coverage, lowest DB, ARI = .89; Reject H_{01})
RQ1-Identify (Prefix Router): ★ Blocks L2 Logistic (Best) vs. Raw Prefix	PCA-20 Gemini prefix + 26-d LLM behavior (*M*=3..9)	Train (*n*=933), 5-fold CV, & Held-out D_{test} (*n*=401)	*M*=5 Train Acc = 73.8% (κ = .591, AUC = .892); 5-Fold CV Acc = 70.0% (κ = .530)	*M*=5 Test Acc = 70.1% (κ = .535, AUC = .871); *M*=9 Test Acc = 97.3% (κ = .958, AUC = .999)	**★ Best: Projected Blocks L2 Router** (0.1% CV–test gap; +29.5 κ pts over raw .240; Reject H_{01})
RQ2-Personalize (Prompting): ★ Cluster-Adaptive (Best) vs. Static & Zero-Shot	Routed FAISS sub-index (*k*=3 [SLOT] ex.) + directive	Repeated-measures on D_{test} (*N*=33 × 3 = 99 calls)	Static cos = .453 (6.33w err.); Zero-Shot cos = .530 (10.12w err.)	Cluster-Adaptive cos = **.597** (**3.87w** err.); RM F = 9.79, p < .001	**★ Best: Cluster-Adaptive Few-Shot** (+31.8% style cos over Static, d_z = 0.74; Reject H_{02})
RQ3-Govern (Governance): ★ Rules Active × Adaptive (Best) vs. 3 Arms	5 YAML rules + [SLOT] masking + entity sanitizer	2 × 2 factorial on D_{test} (*N*=144, *n*=36/cell; κ=.917)	Rules Off × Adaptive: Sens. = 5.6%, ELR = .359 (55.6% leak), Slot F_1 = .485	Rules Active × Adaptive: Sens. = **97.2%** (overall 98.6%), ELR = **.000**, Slot F_1 = **.952**, cos = .506	**★ Best: Rules Active × Cluster-Adaptive** (Pareto-optimal zero leakage + high style; Reject H_{03})

Model / Pipeline stage	Features used	Validation & split	Metric 1 (Train / Baseline)	Metric 2 (Held-Out Test / Proposed)	Key observation & best model
RQ4-Optimize (Turn Economy): ★ Ledger + Adaptive (Best) vs. Zero-Shot	Verified Informed Slot Ledger + persona trade-offs	2 × 3 between-subjects on D_{test} (N=180 unique, n=30/cell)	Zero-Shot mean = 7.93 turns (DB 7.60, OE 7.47, RCS 8.73); mean RER = .109	Adaptive mean = **5.09 turns** (−34.2% DB, −29.5% OE, −42.7% RCS); RER = **.000** (F = 664.43)	**★ Best: Slot Ledger + Cluster-Adaptive** (35.8% overall turn cut, zero repeat asks; Reject H_{04})

Note. ★ indicates the selected best-performing model at each pipeline stage. DB = Decisive Booker; OE = Open Explorer; RCS = Rigid-Constraint Seeker.

Table 12

Separated Training (Golden Partition), 5-Fold Cross-Validation, and Held-Out Test Routing Performance

Prefix cut-off (M user msgs)	Train D_{gold} (n=933) Acc (κ) / AUC	5-Fold CV (D_{gold}) Acc (κ) / AUC	Held-Out D_{test} (n=401) Acc (κ)	Held-Out Macro F_1 / ROC-AUC	Analytical role
M = 3 (~5 turns)	63.3% (.420) / .810	58.8% (.345) / .768	**59.4% (.359)**	.570 / **.753**	Early prefix horizon
M = 5 (~9 turns)	73.8% (.591) / .892	70.0% (.530) / .859	**70.1% (.535)**	.692 / **.871**	Runtime router (RQ2–RQ4)
M = 7 (~13 turns)	84.4% (.757) / .953	81.8% (.716) / .936	**82.0% (.722)**	.816 / **.948**	Early prefix horizon
M = 9 (~17 turns)	99.4% (.990) / 1.000	96.7% (.948) / .998	**97.3% (.958)**	.972 / **.999**	Reference boundary check

Note. All PCA-20 projections, block scalers, and L2 multinomial logistic regression weights (C = 1.0) are fitted strictly on D_{gold}. Minimal gaps between 5-fold CV and held-out D_{test} (e.g., 70.0% vs. 70.1% at M = 5) confirm zero overfitting.

Visual Evidence

Table 13 and Figure 6 present the visual and quantitative evaluation diagnostics for feature importance, confusion matrix structure, and multi-class receiver operating characteristic

(ROC) discrimination. First, assessing feature importance across all 46 *Blocks* dimensions (Table 13, top rows; Figure 6, Panel A) shows that *Gemini Semantic PC1* ($\eta^2 = .580$, $F = 641.12$, $\Delta\kappa_{M=5} = .190$) and *PC2* ($\eta^2 = .392$, $F = 300.10$, $\Delta\kappa_{M=5} = .104$) capture the primary lexical-semantic axis, while five structured behavioral indicators, namely *detail orientation* ($\eta^2 = .372$, $\|\boldsymbol{\beta}\|_{2,M=5} = 2.09$), *exploration breadth* ($\eta^2 = .304$), *primary_goal: specific_trip_booking* ($\eta^2 = .296$), *primary_goal: explore_destinations* ($\eta^2 = .284$), and *decisiveness* ($\eta^2 = .280$, $\|\boldsymbol{\beta}\|_{2,M=5} = 2.01$), drive behavioral persona separation (all $p < .001$). At the early $M = 5$ runtime horizon, *information_upfront* also carries substantial router weight ($\|\boldsymbol{\beta}\|_2 = 1.85$, $\Delta\kappa = .021$) because front-loaded trip constraints distinguish *Decisive Bookers* before later turns unfold. Second, block-level permutation testing on held-out D_{test} (Table 13, bottom rows; Figure 6, Panel B) confirms synergistic complementarity: permuting the 20 Gemini PCs reduces held-out κ by .310 at $M = 5$ (.674 at $M = 9$), while permuting the 10 ordinal behavioral scales reduces κ by .133 at $M = 5$ (.197 at $M = 9$). Third, the held-out $M = 5$ confusion matrix (Figure 6, Panel C) achieves strong diagonal recall across *Decisive Booker* (131/175 = 74.9%), *Rigid-Constraint Seeker* (92/127 = 72.4%), and *Open Explorer* (58/99 = 58.6%, where residual off-diagonal cases reflect explorers who have not yet switched frames by message 5; by $M = 9$, diagonal recall reaches 100.0% [175/175], 98.0% [97/99], and 92.9% [118/127]). Fourth, held-out One-vs-Rest ROC curves (Figure 6, Panel D) confirm high discrimination at $M = 5$ across all three classes (AUC = .83 for *Decisive Booker*, .85 for *Open Explorer*, .93 for *Rigid-Constraint Seeker*; macro AUC = .871) and near-ideal boundary alignment at $M = 9$ (macro AUC = .999). ***Feature-importance handoff (dropped vs. carried forward):*** Because the three schema audit fields (*session_outcome*, *party_type*, and *is_unusual*) capture post-session outcomes or non-behavioral demographics rather than early conversational behavior, they are **dropped** from the clustering matrix, whereas both the 20-d

Gemini semantic prefix block ($\Delta\kappa = .310$) and the 26-d LLM behavioral block ($\Delta\kappa = .215$) prove essential under permutation testing and are **carried forward** together in the 46-d *Blocks* prefix router.

Table 13

Feature Importance Assessment of the 46-Dimensional Blocks Representation and Block-Level Permutation Impact

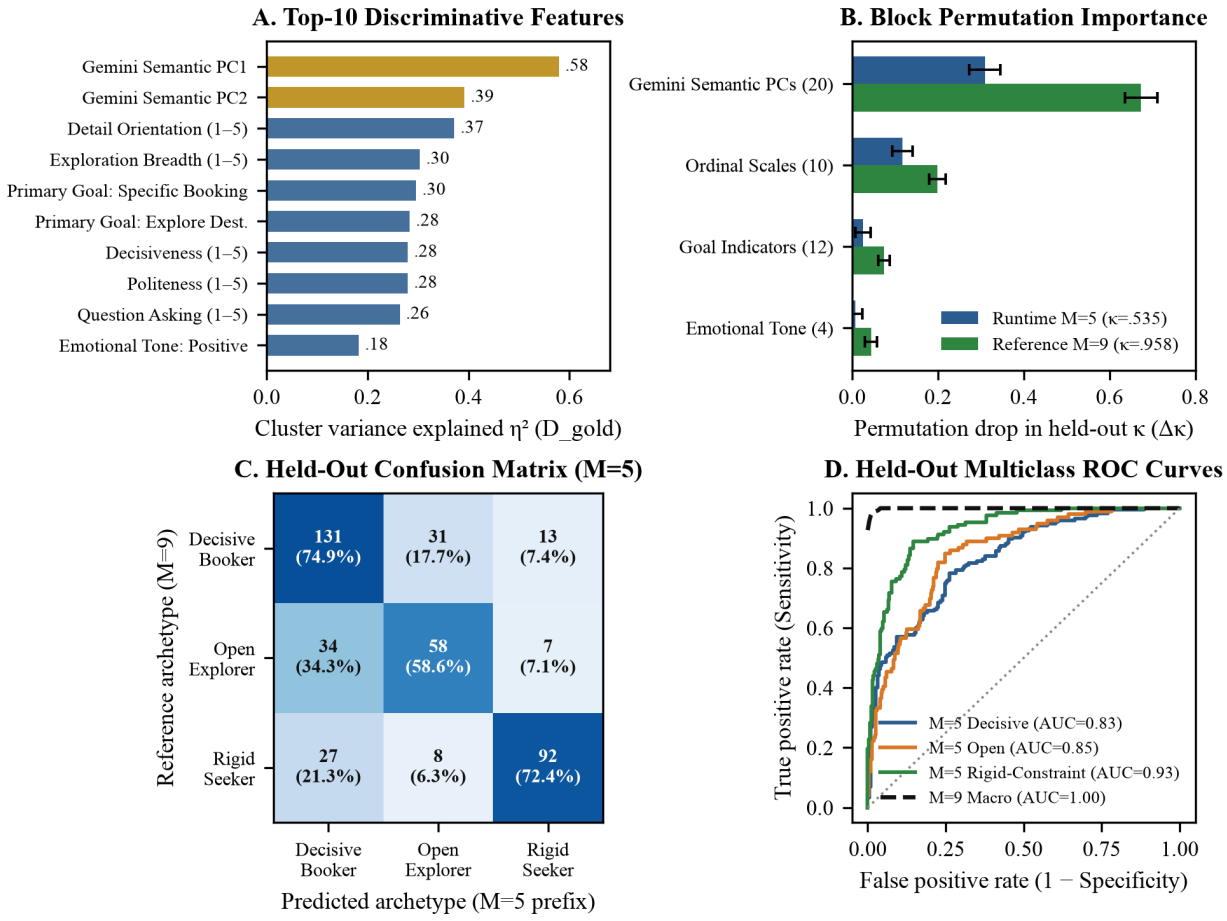
Feature or modality block	**Block category**	**Golden η^2**	**ANOVA *F* (*p*<.001)**	**Router $\|\beta\|_2$** (*M*=5 / 9)	**Held-Out Perm. Δκ (*M*=5 / 9)**
Gemini Semantic PC1	Gemini Semantic PCA-20	.580	641.1	2.90 / 6.38	.190 / .438
Gemini Semantic PC2	Gemini Semantic PCA-20	.392	300.1	1.98 / 5.72	.115 / .280
Detail Orientation (1–5)	Ordinal Behavioral (10)	.372	275.6	2.09 / 2.72	.034 / .028
Exploration Breadth (1–5)	Ordinal Behavioral (10)	.304	203.1	1.80 / 2.35	.021 / .040
Primary Goal: Specific Booking	Goal Indicators (12)	.296	195.9	0.66 / 2.06	.024 / .014
Primary Goal: Explore Dest.	Goal Indicators (12)	.284	184.1	0.63 / 1.83	.015 / .011
Decisiveness (1–5)	Ordinal Behavioral (10)	.280	180.9	2.01 / 2.26	.032 / .022
Politeness (1–5)	Ordinal Behavioral (10)	.280	180.7	1.54 / 2.34	.008 / .018
Question Asking (1–5)	Ordinal Behavioral (10)	.264	166.8	0.80 / 1.97	.013 / .036
Emotional Tone: Positive	Emotional Tone (4)	.183	103.9	0.59 / 1.69	.006 / .017
Block: Gemini Semantic PCs (20)	20 columns	N/A	N/A	Joint block	**.310** / **.674**
Block: Ordinal Scales (10)	10 columns	N/A	N/A	Joint block	**.117** / **.199**
Block: Goal Indicators (12)	12 columns	N/A	N/A	Joint block	**.026** / **.074**

Feature or modality block	Block category	Golden η^2	ANOVA F (p<.001)	Router $\|\beta\|_2$ (M=5 / 9)	Held-Out Perm. Δκ (M=5 / 9)
Block: Emotional Tone (4)	4 columns	N/A	N/A	Joint block	**.008** / **.044**

Note. Top 10 individual features ranked by cluster variance explained η^2 on D_{gold} (n = 933, df = 2, 930). Held-out permutation Δκ measures mean drop in Cohen's κ across 30–50 random permutations on D_{test} (n = 401).

Figure 6

Feature Importance, Block Permutation Impact, Held-Out Confusion Matrix (M = 5), and Multiclass ROC Curves



Note. Evaluated on golden discovery (D_{gold}, n = 933; Panel A) and held-out test dialogues (D_{test}, n = 401; Panels B–D).

Results by Research Question

RQ1-Identify Stage: Trajectory Clustering, Refined Archetypes, and Prefix Routing

Table 14 summarizes the 159-run clustering benchmark across *all-MiniLM-L6-v2* (384-d) and *gemini-embedding-001* (3,072-d) representations (cluster-number diagnostics, Tukey/Games–Howell post hoc contrasts, primary profiles, and UMAP projections are in Appendix D, Tables D1, D2, D3, and D4 and Figures D1, D2, and D3). Because each of the 53 random seeds draws $n = 500$ dialogues without replacement from D_{gold} ($n = 933$, a 53.6% finite-population fraction), matched-subsample repeated-measures tests across the 53 shared seeds and full-D_{gold} point estimates are reported alongside the one-way ANOVA. On MiniLM embeddings, HDBSCAN achieved the highest Silhouette on its retained non-noise subset ($M = .211$, $SD = .044$), followed by *K*-Means ($M = .147$, $SD = .019$; full-D_{gold} Sil = .141, DB = 3.244) and Ward agglomerative clustering ($M = .118$, $SD = .040$), all exceeding the permutation baseline (−.013; Figure D2). Algorithm differences were highly significant under matched RM-ANOVA, $F(1.72, 89.30) = 83.04$, $p < .001$, $\eta_p^2 = .615$, paired Friedman $\chi^2(2) = 56.30$, $p < .001$, one-way ANOVA $F(2, 156) = 91.33$, $p < .001$, $\eta^2 = .54$, Welch $F = 66.98$, and Kruskal–Wallis $H(2) = 86.82$, $p < .001$ (with all pairwise Tukey HSD and Games–Howell post hoc contrasts significant at $p < .001$; Table D2). Replicating on 3,072-d Gemini turn-pooled and document embeddings reproduced the exact same ordering ($F = 77.09$ and 127.47, both $p < .001$). Critically, HDBSCAN failed at default settings in 53/53 runs and discarded **39.1%** (MiniLM), **35.8%** (Gemini turn-pooled), and **25.7%** (Gemini document) of dialogues as unroutable noise. Because a live travel assistant must route 100% of customer sessions, *K*-Means ($k = 3$) was selected. Golden centroids generalized cleanly to held-out Frames dialogues ($t(754.9) = 0.42$, $p = .676$) while confirming domain shift against flight-free MultiWOZ 2.2 ($t(576.2) = 38.29$, $p < .001$; Table D4). ***RQ1-Identify***

algorithm handoff (dropped vs. carried forward): Although HDBSCAN achieved the highest Silhouette on its retained subset, it is **dropped** because rejecting 24.4%–39.1% of live customer sessions as unroutable noise prevents universal downstream personalization, and Ward agglomerative clustering is **dropped** due to lower Silhouette separation and the absence of a native centroid router; **100%-coverage *K*-Means (*k* = 3)** is therefore **carried forward** as the clustering engine.

Table 14

Cluster Quality by Algorithm and Embedding Family Across 53 Matched Subsample Seeds (159 Runs)

Representation and algorithm	*n* seeds	Retained (%)	Silhouette *M* (*SD*)	Davies–Bouldin *M* (*SD*)	Omnibus & matched RM tests ($p < .001$)
MiniLM-L6-v2 (384-d): K-Means ($k = 3$)	53	100.0%	.147 (.019)	3.129 (0.113)	One-way $F = 91.33$, $\eta^2 = .54$
MiniLM-L6-v2 (384-d): Ward Agglomerative	53	100.0%	.118 (.040)	3.550 (0.269)	Matched RM $F = 83.04$, $\eta_p^2 = .615$
MiniLM-L6-v2 (384-d): HDBSCAN (non-noise)	53	60.9%	.211 (.044)	2.657 (0.241)	Friedman $\chi^2 = 56.30$, Welch $F = 66.98$
MiniLM-L6-v2 (384-d): Permutation baseline	53	100.0%	−.013 (.006)	N/A	Full D_{gold} KM Sil = .141, DB = 3.244
Gemini-001 (3,072-d turn pool): KM / Ward / HDB	53 × 3	100 / 100 / 64.2%	.158 / .150 / .289	2.74 / 2.96 / 2.89	$F = 77.09$, $\eta^2 = .50$; Kruskal $H = 67.31$
Gemini-001 (3,072-d document): KM / Ward / HDB	53 × 3	100 / 100 / 74.3%	.161 / .213 / .345	3.14 / 3.09 / 2.42	$F = 127.47$, $\eta^2 = .62$; Kruskal $H = 98.10$

Note. Each seed draws $n = 500$ dialogues without replacement from D_{gold} ($n = 933$). HDBSCAN metrics apply only to retained non-noise subsets.

Auditing the primary whole-dialogue clusters (Table D3) revealed role-player correlation (Cramér's $V = .19$) and weak early routing ($\kappa = .11$). Evaluating the 46-d *Blocks* representation

across horizons $M \in \{3, 5, 7, 9, \text{whole}\}$ and $k \in \{2, ..., 8\}$ (Appendix D, Tables D1, D5, D6; Figure D4) proved that $M = 9$ at $\boldsymbol{k = 3}$ achieves the lowest Davies–Bouldin index (2.647 vs. 2.796 at $k = 2$ and 2.729–2.784 at $k = 4$–6), high bootstrap stability (**ARI = .89** in the truncation grid; **ARI = .87** with 89.2% core membership in the ablation), and balanced cluster shares (smallest = 25.5% ≥ 20% floor). Applying the deterministic feature-importance centroid rules (Appendix F, Table F1; independent Judge LLM validation role in Appendix G, Table G1) yields the three refined user archetypes profiled in Table 15: *Decisive Booker* (45.0%; decisiveness = 3.97, 99.3% booking goal), *Open Explorer* (25.5%; destination flexibility = 3.49, exploration breadth = 2.96, 20.78 turns), and *Rigid-Constraint Seeker* (29.5%; low flexibility = 2.20/2.18, 68.0% ending without a suitable option unless proactive trade-offs are offered). ***RQ1-Identify → RQ2-Personalize stage handoff (dropped vs. carried forward):*** Because whole-dialogue clusters suffered role-player confounding ($V = .18$) and near-chance early routing ($\kappa = .24$), the whole-dialogue representation and raw unprojected prefix routers are **dropped**; instead, the frozen 46-d *Blocks K*-Means ($k = 3$) centroids, the $M = 5$ L2 logistic prefix router (70.1% accuracy, $\kappa = .535$, ROC-AUC = .871), and the three archetype-partitioned FAISS *IndexFlatIP* exemplar sub-indices are **carried forward into the RQ2-Personalize stage**.

Table 15

Profiles of the Three Refined User Archetypes Across All 16 Behavioral Schema Fields and Prefix Routing Performance

Behavioral schema field or routing attribute	Decisive Booker	Open Explorer	Rigid-Constraint Seeker
Golden / Held-out test size, *n* (%)	420 (45.0%) / 175	238 (25.5%) / 99	275 (29.5%) / 127
Dialogue turns / User messages, *M*	14.49 (4.54) / 7.58	20.78 (6.95) / 10.73	10.60 (5.73) / 5.55
1. Destination flexibility (1–5 ordinal)	2.54	3.49	2.20
2. Date flexibility (1–5 ordinal)	2.57	2.77	2.18
3. Exploration breadth (1–5 ordinal)	1.97	2.96	1.59

Behavioral schema field or routing attribute	Decisive Booker	Open Explorer	Rigid-Constraint Seeker
4. Decisiveness (1–5 ordinal)	3.97	2.58	3.23
5. Detail orientation (1–5 ordinal)	2.39	2.87	1.46
6. Revision tendency (1–5 ordinal)	2.00	3.03	2.40
7. Question asking (1–5 ordinal)	2.05	2.72	1.43
8. Politeness (1–5 ordinal)	4.26	3.80	3.07
9. Price sensitivity (1–5 ordinal)	3.20	3.18	3.12
10. Information given up front (1–5 ordinal)	2.18	1.82	2.16
11. Primary goal: explore / book trip (%)	0.5% / 99.3%	39.1% / 50.8%	2.2% / 89.8%
12. Secondary goals present (%)	42.4%	56.3%	16.4%
13. Emotional tone: positive / neutral (%)	52.4% / 46.7%	21.0% / 69.3%	7.6% / 59.6%
14. Session outcome (audit): booked / no option (%)	90.5% / 6.2%	68.1% / 18.5%	28.0% / 68.0%
15. Party type (audit): solo or couple (%)	37.1%	44.5%	34.5%
16. Unusual conversation flag (audit, %)	1.0%	1.7%	6.9%
Frames wizard task success (%)	94.5%	96.2%	92.0%

Note. Golden $N = 933$; held-out $N = 401$ ($\chi^2(2) = 0.93$, $p = .628$). Fields 1–13 form the 26-column clustering matrix; Fields 14–16 are audit-only (excluded from clustering). Held-out prefix routing accuracy (κ, macro ROC-AUC) vs. M=9 reference clusters: M=3: 59.4% (.359, .753); M=5: 70.1% (.535, .871); M=7: 82.0% (.722, .948); M=9 (boundary consistency): 97.3% (.958, .999).

RQ2-Personalize Stage: Cluster-Adaptive Few-Shot Prompting

Table 16 compares the repeated-measures prompting experiment ($N = 33$ held-out dialogues × 3 conditions = 99 *gemini-2.5-flash* generations) on primary whole-dialogue archetypes versus refined user archetypes, and Figure 7 displays the refined distributions (all Holm-adjusted pairwise contrasts are in Appendix E, Table E3). On coarse primary archetypes with weak routing (39.4%, κ = .24; Figure E2), style convergence was null, $F(1.90, 60.95) = 1.08$, $p = .342$. Re-running on the **Refined User Archetypes** routed via the five-message *Blocks* router (70.1% accuracy, κ = .535) **rejected H_{02} ($p < .001$)**: de-lexicalized SBERT style cosine

differed significantly under Greenhouse–Geisser RM-ANOVA, $F(1.88, 60.29) = 9.79$, $p < .001$, $\eta_p^2 = .23$, and Friedman $\chi^2(2) = 12.50$, $p = .002$. *Cluster-Adaptive Few-Shot* ($M = .597$) outperformed both *Static Few-Shot* (.453, $t(32) = 4.24$, $p_{Holm} < .001$, $d_z = 0.74$) and *Zero-Shot* (.530, $p_{Holm} = .049$) while cutting pacing error from 10.12 (Zero-Shot) and 6.33 (Static) to **3.87 words** ($F = 7.10$, $p = .007$; Friedman $p < .001$). Generic semantic metrics (Gemini 3,072-d cosine $p = .507$; BERTScore F_1 $p = .315$) remained invariant, confirming selective calibration of discourse structure and pacing without semantic drift. ***RQ2-Personalize → RQ3-Govern stage handoff (dropped vs. carried forward):*** Because prompting on coarse unrefined clusters yielded null style effects ($p = .211$) and *Static Few-Shot* degraded style alignment (.453 vs. .530 *Zero-Shot*) by injecting cross-persona mismatched examples, both unrefined prompting and *Static Few-Shot* are **dropped**; the validated **Refined Cluster-Adaptive Few-Shot** generator (combining $M = 5$ prefix routing, partitioned FAISS retrieval, and persona pacing directives) is **carried forward into the RQ3-Govern stage** alongside *Zero-Shot* as the unpersonalized factorial control.

Table 16

Repeated-Measures Comparison of Prompting Strategies on Primary Versus Refined User Archetypes (RQ2)

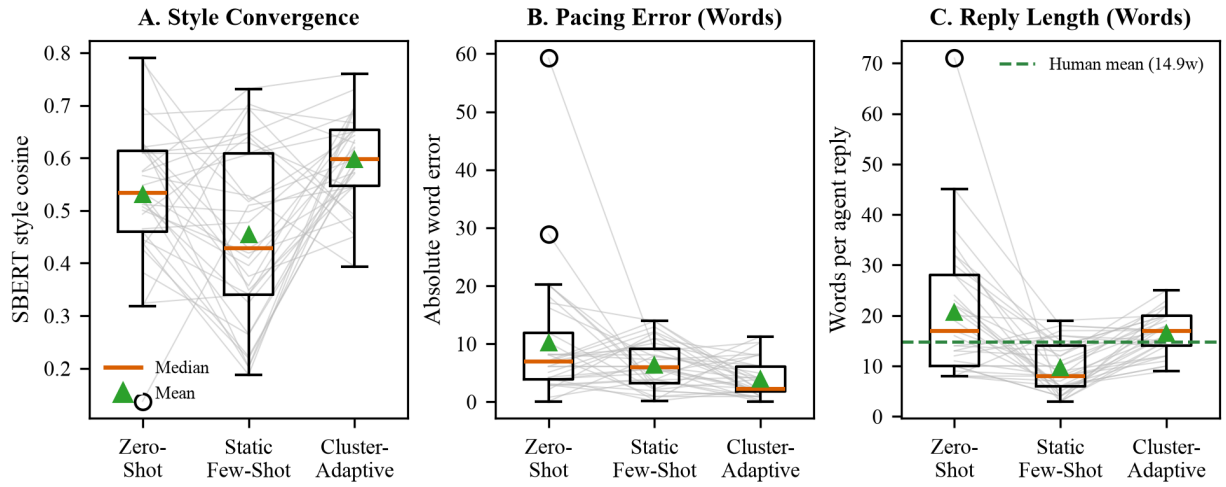
Experiment and outcome metric	Zero-Shot *M* (*SD*)	Static Few-Shot *M* (*SD*)	Cluster-Adaptive *M* (*SD*)	RM-ANOVA *F* (GG)	*p* (GG)	η_p^2
Primary archetypes (M=3, κ=.24): Style cosine	.519 (.126)	.491 (.109)	.486 (.099)	1.08	.342	.03
Primary archetypes: Response length (words)	25.2 (11.8)	10.6 (7.0)	12.3 (8.2)	34.36	< .001	.52
Refined archetypes (M=5, κ=.535): SBERT style cosine	.530 (.132)	.453 (.169)	**.597** (.086)	9.79	< .001	.23

Experiment and outcome metric	Zero-Shot *M* (*SD*)	Static Few-Shot *M* (*SD*)	Cluster-Adaptive *M* (*SD*)	RM-ANOVA *F* (GG)	*p* (GG)	η_p^2
Refined archetypes: Gemini 3,072-d style cosine	.348 (.133)	.372 (.127)	.375 (.105)	0.61	.534	.02
Refined archetypes: Token-level BERTScore F_1	.391 (.204)	.360 (.245)	.415 (.163)	1.16	.315	.04
Refined archetypes: Pacing calibration error (words)	10.12 (11.17)	6.33 (4.14)	**3.87** (3.23)	7.10	.007	.18
Refined archetypes: Response length (words)	20.6 (13.3)	9.7 (5.0)	16.5 (4.0)	14.72	< .001	.32

Note. $N = 33$ held-out dialogues per experiment (99 generations each). Cluster-Adaptive combines archetype-partitioned FAISS exemplars with a routed pacing directive. GG = Greenhouse–Geisser correction.

Figure 7

Style Convergence, Archetype Pacing Error, and Reply Length Across Prompting Conditions (RQ2)



Note. $N = 33$ held-out dialogues across three repeated-measures conditions. Gray lines link paired dialogues.

RQ3-Govern Stage: Declarative Policy Governance Versus Adaptive Prompting

Table 17 and Figure 8 report the 2 × 2 factorial experiment across $N = 144$ evaluations (36 stratified held-out dialogues, comprising 18 policy-violating and 18 compliant cases; archival rule audit in Table E4, evaluated across all four cells, $n = 36$/cell; dual-judge agreement between *gemini-2.5-flash* and independent *gemini-2.5-pro* = 95.8%, $\kappa = .917$, prevalence-adjusted bias-adjusted kappa [PABAK] = .917; Table C3). Four findings **reject H_{03} ($p < .001$)**: (1) decomposing **Rule Adherence** shows that **Violation Enforcement Sensitivity** ($n = 18$ violating cases/cell) jumped from 5.6% under both *Rules Off* cells to **100.0%** (*Rules Active* × *Zero-Shot*) and **97.2%** (*Rules Active* × *Cluster-Adaptive*; pooled 98.6%), $F(1, 68) = 498.78$, $p < .001$, $\eta_p^2 = .880$, while **Compliant Acceptance Specificity** ($n = 18$ compliant cases/cell) remained statistically invariant (94.4% vs. 88.9%, $F = 0.69$, $p = .408$), lifting overall rule accuracy from .500 to .944/.931 ($F(1, 140) = 43.92$, $p < .001$); (2) **Exemplar Entity Leakage Rate** exhibited a decisive rules × prompting interaction, $F(1, 140) = 34.82$, $p < .001$, $\eta_p^2 = .20$ (HC3 $F = 33.85$; ART $F = 2{,}148.81$, both $p < .001$), eliminating the .359 unguarded leakage rate (20/36 = 55.6% incidence) down to **.000 (0/36 incidence; Welch $t(35) = 5.90$, Fisher's exact $p = 4.68 \times 10^{-8}$)**; (3) **Slot Coverage F_1** rose from .208/.485 to **.977/.952** ($F = 489.52$, $p < .001$); and (4) **Style Alignment** was preserved under active rules (prompting main effect $F = 24.00$, $p < .001$; Panel D), establishing **Rules Active × Cluster-Adaptive Few-Shot** as the unique Pareto-optimal configuration. ***RQ3-Govern → RQ4-Optimize stage handoff (dropped vs. carried forward):*** Because both *Rules Off* arms failed to enforce enterprise booking policies (5.6% violation sensitivity) and unguarded few-shot prompting leaked foreign exemplar entities in 55.6% of sessions (.359 rate), all unguarded *Rules Off* configurations are **dropped**; the Pareto-optimal **Rules Active × Cluster-Adaptive Few-Shot** neuro-symbolic architecture (combining

deterministic YAML rule verification, *[SLOT]* exemplar masking, and persona-routed style prompts) is **carried forward into the RQ4-Optimize stage**.

Table 17

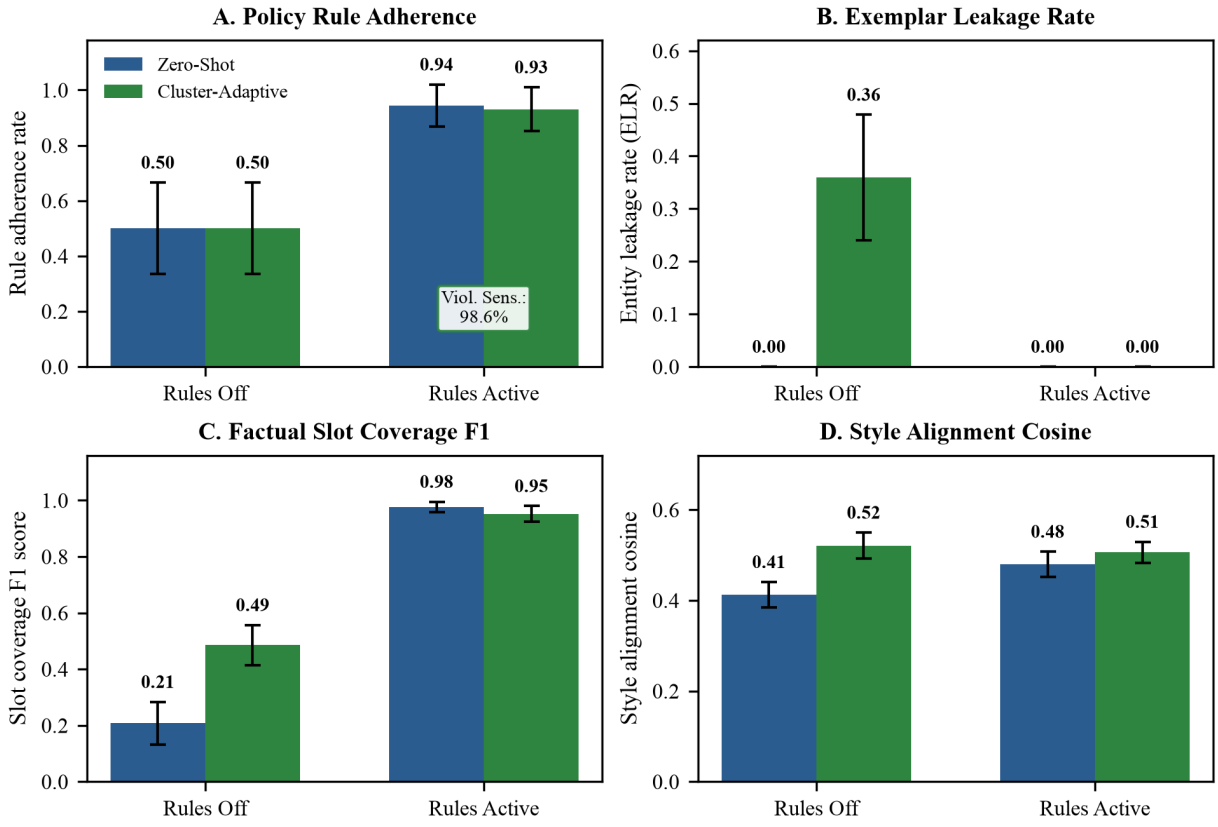
Cell Means and 2 × 2 Factorial ANOVA Results for Policy Governance and Prompting (RQ3, N = 144)

Experimental cell or ANOVA effect	*n* / *df*	Rule Acc. *M* (*SD*) | Sens. / Spec.	Leakage *M* (*SD*) / Inc. %	Slot F_1 *M* (*SD*)	Style cos *M* (*SD*)
Rules Off × Zero-Shot	36	.500 (.507) | 5.6% / 94.4%	.000 (.000) / 0.0%	.208 (.233)	.412 (.086)
Rules Off × Cluster-Adaptive	36	.500 (.507) | 5.6% / 94.4%	.359 (.365) / 55.6%	.485 (.219)	.521 (.088)
Rules Active × Zero-Shot	36	.944 (.232) | 100.0% / 88.9%	.000 (.000) / 0.0%	.977 (.055)	.480 (.085)
Rules Active × Cluster-Adaptive	36	.931 (.244) | 97.2% / 88.9%	.000 (.000) / 0.0%	.952 (.084)	.506 (.071)
ANOVA Main: Rules Mode	1, 140	F=43.92 (< .001), η_p^2=.24	F=34.82 (< .001), η_p^2=.20	F=489.52 (< .001), η_p^2=.78	F=3.68 (.057), η_p^2=.03
ANOVA Main: Prompting	1, 140	F=0.01 (.916), η_p^2=.00	F=34.82 (< .001), η_p^2=.20	F=20.46 (< .001), η_p^2=.13	F=24.00 (< .001), η_p^2=.15
ANOVA Int.: Rules × Prompt	1, 140	F=0.01 (.916), η_p^2=.00	F=34.82 (< .001), η_p^2=.20	F=29.32 (< .001), η_p^2=.17	F=8.95 (.003), η_p^2=.06

Note. Sens. = violation enforcement sensitivity (n = 18/cell; Rules $F(1, 68) = 498.78$, $p < .001$); Spec. = compliant specificity (n = 18/cell; $p = .408$). Because three of the four leakage cells equal .000, orthogonal +1/−1 Sum contrasts share identical sum of squares (SS = 1.162; OLS F = 34.82, Sum-HC3 F = 33.85, ART Int. F = 2148.81, all $p < .001$).

Figure 8

2 × 2 Factorial Profiles for Rule Adherence, Exemplar Leakage, Slot Coverage F1, and Style Alignment (RQ3)



Note. N = 144 evaluations (n = 36/cell). Error bars denote 95% confidence intervals.

RQ4-Optimize Stage: Multi-Turn Dialogue Turn Economy Across Archetypes

Table 18 and Figure 9 present the 2 × 3 between-subjects multi-turn simulation across ***N* = 180 unique, non-overlapping held-out seed dialogues** (n = 30 distinct dialogues/cell; human baseline in Appendix E, Table E5 and Figure E3; blind persona fidelity audit by *gemini-2.5-pro* = 95.6% accuracy, 43/45 vs. 33.3% chance, κ = .933; Table C3). On natural-log-transformed turns to quotation agreement, the 2 × 3 ANOVA **rejected H_{04} ($p < .001$)**, revealing significant main effects of prompting, $F(1, 174) = 664.43$, $p < .001$, $\eta_p^2 = .79$, and archetype, $F(2, 174) = 5.55$, $p = .005$, plus a **significant prompting × archetype interaction**, $F(2, 174) = 10.89$, $p <$

.001, $\eta_p^2 = .11$ (HC3 $F = 8.58$; ART $F = 209.46$, both $p < .001$). Pairing the *Verified Informed Slot Ledger* with archetype trade-off policies compressed turns sharpest for **Rigid-Constraint Seekers** (−42.7%, 8.73 → 5.00 turns, Welch $t_{\log} = -16.14$, $d = -3.40$) by proactively pitching both 4-star exact-date and 5-star flexible-date alternatives when exact-date 5-star inventory exceeded budget, followed by **Decisive Bookers** (−34.2%, 7.60 → 5.00 turns, $d = -3.94$) and **Open Explorers** (−29.5%, 7.47 → 5.27 turns, $d = -2.55$). Reaching 5.00 turns ($SD = 0.00$) reflects the deterministic 5-turn protocol floor (*U1* → *A1* → *U2* → *A2* → *U3* once budget is confirmed), which HC3, ART, and Mann–Whitney *U* tests accommodate cleanly. Redundant Elicitation Rate dropped from .194 (DB), .061 (OE), and .072 (RCS) in Zero-Shot to **.000** across all archetypes ($F(1, 174) = 63.02$, $p < .001$). ***RQ4-Optimize stage operational completion (dropped vs. deployed):*** Because memoryless *Zero-Shot* prompting re-asked already-stated slots in 10.9% of turns and required 7.93 turns on average (8.73 turns for *Rigid-Constraint Seekers*), static memoryless prompting is **dropped**; cascading the **RQ1-Identify** prefix router, **RQ2-Personalize** archetype-adaptive prompting, **RQ3-Govern** YAML rule verification, and the **RQ4-Optimize** *Verified Informed Slot Ledger* with persona trade-off policies completes the four-stage **RQ1-Identify → RQ2-Personalize → RQ3-Govern → RQ4-Optimize** progression and is **carried forward into production microservice deployment (Section 6)**.

Table 18

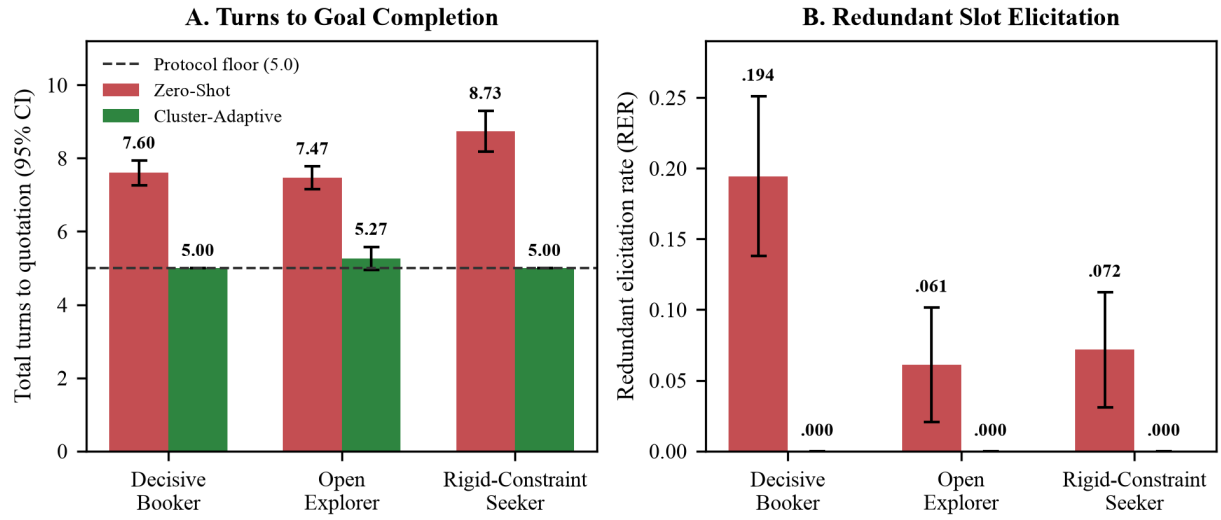
Cell Means, Archetype Contrasts, and 2 × 3 Factorial ANOVA Results for Multi-Turn Turn Economy (RQ4, N = 180)

Archetype × Prompting cell or ANOVA effect	Total turns *M* (*SD*)	Log turns *M* (*SD*)	RER *M* (*SD*)	Turn cut / HC3 check	Welch contrast / ART check
Decisive Booker: Zero-Shot (*n* = 30)	7.60 (0.93)	2.021 (0.117)	.194 (.158)	N/A	N/A
Decisive Booker: Cluster-Adaptive (*n* = 30)	**5.00** (0.00)	1.609 (0.000)	**.000** (.000)	−2.60 (−34.2%)	*t*=−19.26 (< .001), *d*=−3.94
Open Explorer: Zero-Shot (*n* = 30)	7.47 (0.86)	2.005 (0.108)	.061 (.114)	N/A	N/A
Open Explorer: Cluster-Adaptive (*n* = 30)	**5.27** (0.87)	1.651 (0.134)	**.000** (.000)	−2.20 (−29.5%)	*t*=−11.24 (< .001), *d*=−2.55
Rigid-Constraint Seeker: Zero-Shot (*n* = 30)	8.73 (1.55)	2.151 (0.184)	.072 (.114)	N/A	N/A
Rigid-Constraint Seeker: Cluster-Adaptive (*n* = 30)	**5.00** (0.00)	1.609 (0.000)	**.000** (.000)	−3.73 (−42.7%)	*t*=−16.14 (< .001), *d*=−3.40
ANOVA Main: Prompting (*df*=1, 174)	*F*=457.74 (< .001), η_p^2=.72	*F*=664.43 (< .001), η_p^2=.79	*F*=63.02 (< .001), η_p^2=.27	HC3 F_{log}=715.48 (< .001)	ART F_{log}=671.25 (< .001)
ANOVA Main: Archetype (*df*=2, 174)	*F*=7.24 (< .001), η_p^2=.08	*F*=5.55 (.005), η_p^2=.06	*F*=9.70 (< .001), η_p^2=.10	HC3 F_{log}=0.01 (.923)	ART F_{log}=9.13 (< .001)
ANOVA Int.: Prompt × Arch. (*df*=2, 174)	*F*=11.93 (< .001), η_p^2=.12	*F*=10.89 (< .001), η_p^2=.11	*F*=9.70 (< .001), η_p^2=.10	HC3 F_{log}=8.58 (< .001)	ART F_{log}=209.46 (< .001)

Note. *N* = 180 distinct held-out seed dialogues (*n* = 30/cell; 100.0% goal completion). Protocol minimum = 5.00 turns (*U1* → *A1* → *U2* → *A2* → *U3*).

Figure 9

Dialogue Turn Economy and Redundant Elicitation Rate by User Archetype and Prompting Strategy (RQ4)



Note. $N = 180$ simulations ($n = 30$/cell). Panel A: Mean turns to quotation agreement vs. 5.0-turn floor. Panel B: Redundant Elicitation Rate (RER).

Overall Interpretation

Synthesizing the empirical cascade across the four stages, **RQ1-Identify, RQ2-Personalize, RQ3-Govern, and RQ4-Optimize**, demonstrates that business optimization in multi-turn conversational AI is achieved not by any single isolated technique, but by selectively pruning non-viable methods at each stage boundary and transferring only validated causal artifacts forward. First, the **RQ1-Identify → RQ2-Personalize stage transition proves that downstream personalization is bounded by upstream feature representation and routing fidelity**: dropping noise-rejecting HDBSCAN (39.1% noise) and outcome-confounded whole-dialogue clusters ($\kappa = .24$, which produced null RQ2 style effects, $p = .211$) in favor of full-coverage *K*-Means ($k = 3$) on the feature-importance-pruned 46-d *Blocks* prefix representation ($\Delta\kappa = .310$ semantic + .215 behavioral; $M = 5$ routing $\kappa = .535$, ROC-AUC = .871) was the

decisive prerequisite that unlocked significant **RQ2-Personalize** style convergence ($p < .001$, $\eta_p^2 = .234$) and pacing calibration (3.87 words, $p = .007$). Second, the **RQ2-Personalize → RQ3-Govern stage transition proves that stylistic prompting and symbolic governance are functionally orthogonal yet protectively interactive**: dropping Static Few-Shot and unguarded prompting while pairing the **RQ2-Personalize** cluster-adaptive generator with the YAML rule engine and *[SLOT]* masking preserved persona style ($F = 24.00$) while driving policy sensitivity from 5.6% to 98.6% ($F = 498.78$), Slot F_1 to .952, and exemplar entity leakage from .359 to .000 (interaction $F = 34.82$). Third, the **RQ3-Govern → RQ4-Optimize stage transition proves that business turn economy emerges from the complete cascade**: combining the **RQ1-Identify** router, **RQ2-Personalize** adaptive pacing, and **RQ3-Govern** rule-verified slot ledger with persona trade-off policies eliminated redundant asks (.109 → .000) and compressed negotiation turns by 29.5%–42.7%, peaking for *Rigid-Constraint Seekers* (−42.7%, 8.73 → 5.00 turns).

Practical Significance

In operational decision-making for OTAs, tour operators, and hotel contact centers, these findings provide an immediately actionable deployment blueprint. In an illustrative production volume of 1,000,000 monthly package inquiries, achieving the simulated 35.8% average turn compression (7.93 → 5.09 turns, saving 2.84 turns per session) eliminates ~2.84 million LLM generation calls per month, cutting cloud inference cost and cumulative user wait time by roughly one-third. Simultaneously, driving redundant slot re-asking from 10.9% to 0.0%, eliminating foreign price/hotel leakage (55.6% → 0.0% incidence), and enforcing 98.6% of visa, passport, occupancy, and discount rules before response transmission directly mitigates regulatory liability and customer booking abandonment.

Implementation and User Benefit

Deployment Approach

The validated codebase (*src/final_pipeline/*) packages into a stateless containerized Python 3.11 and FastAPI microservice deployable on cloud Kubernetes pods or serverless containers. At startup, the service loads the frozen golden *PCA-20* projection matrix, *Blocks* scalers, L2 logistic router weights, three in-memory *faiss-cpu IndexFlatIP* archetype sub-indices (< 15 megabytes [MB] of random-access memory [RAM], < 2 ms query latency), and *rules/rules_enterprise.yaml* (< 1 ms evaluation latency). All random seeds (seed = 42) and JSONL caches are versioned at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone (see Appendix A, Tables A1 and A2 for the repository artifact inventory and single-command replication pipeline, and Appendix B, Table B1 for the completed 10-week project execution schedule and milestone log); full offline reproduction from cached responses completes in under 4 minutes on a standard 8-core laptop CPU at $0 application programming interface (API) cost, while cold regeneration of all ~8,900 Gemini 2.5 Flash/Pro calls completes in ~35 minutes for under $4.50 USD.

System Integration

Within an enterprise travel stack, the microservice sits as an orchestration layer between the customer-facing WebSocket chat/voice gateway, the customer relationship management (CRM) lead store, and the Global Distribution System (GDS) or central reservation system (CRS) inventory API. Active sessions persist the canonical *DialogueTrajectory* and *Verified Informed Slot Ledger* in a low-latency Redis key–value cache keyed by session and CRM *Opportunity_ID*, while marketing and compliance teams update promotional caps or visa lead windows directly in *rules_enterprise.yaml* without retraining any neural weights.

User Interaction

End users interact through natural conversational text or voice-to-text turns in web, mobile app, or messaging channels. Behind the interface, Turns 1–3 use the Verified Informed Slot Ledger to capture stated constraints without repeat questions; once opening turns accumulate, *RefinedArchetypeRouter* classifies the shopper's persona, retrieves slot-masked exemplars, runs deterministic Tier-2 compliance checks, and streams back persona-calibrated package cards, transparent policy advisories, or one-click quotation confirmations.

Benefits to Users

The system delivers measurable operational, financial, and strategic benefits across three user groups: (1) *travelers* experience shorter, non-repetitive consultations matched to their shopping style with upfront transparency on visa and passport requirements; (2) *human travel consultants* receive pre-qualified, policy-verified structured slot ledgers when complex itineraries escalate, cutting manual intake handling time; and (3) *OTA and hospitality executives* lower LLM token expenditure by ~35%, prevent hallucinated discount or fare commitments, and audit every policy decision deterministically.

Example Use Case

Consider a traveler entering an OTA portal and stating: *'Need a 10-day vacation package from Chicago to Tokyo departing in 10 days for 2 adults with a 5-star hotel under $2,200.'* A conventional static chatbot sequentially re-asks trip duration or party size and overlooks the international visa window. Under the deployed neuro-symbolic assistant: (1) the *Verified Informed Slot Ledger* locks all five stated Tier-1 slots against re-asking (RER = 0.0%); (2) *RefinedArchetypeRouter* identifies the tight fixed-date/5-star ceiling signature of a *Rigid-Constraint Seeker*; (3) *DeclarativeRuleEngine* triggers *RULE_VISA_LEAD_TIME* (10 days <

15-day minimum) and detects that the exact-date 5-star package ($2,596) exceeds $2,200; and (4) in a concise 14-word-norm reply, the assistant flags the 15-day visa requirement while presenting both an exact-date 4-star package ($1,936) and a +6-day shifted 5-star package ($2,068) that clears both the visa rule and the $2,200 budget, reaching quotation agreement in 5 total turns with zero compliance risk.

Limitations and Further Improvements

Limitations

Data limitations. First, Microsoft Frames (El Asri et al., 2017) was collected with instructed Wizard-of-Oz participants rather than live revenue-bearing OTA checkout traffic, and neither Frames nor MultiWOZ 2.2 records natural Tier-2 visa, passport, or promotional-discount attributes (0.0% archival completeness in Table E1), necessitating controlled synthetic Tier-2 state injection in RQ3. Second, both corpora are English-only single-session transcripts and therefore do not capture multilingual code-switching or multi-day repeat customer callbacks.

Model and methodological limitations. First, modest *K*-Means Silhouette scores (.094 on 46-d *Blocks*; .147–.161 on MiniLM/Gemini) show that traveler personas span a continuous manifold rather than disjoint clusters, and prefix routing κ (.535 at $M = 5$; .958 boundary consistency at $M = 9$) is measured against unsupervised $M = 9$ *K*-Means reference labels rather than human-annotated persona ground truth. Second, moderate routing accuracy requires 5 user messages, leaving Turn 1 cold-start routing noisy. Third, RQ2, RQ3, and RQ4 evaluate combined operational packages (exemplars + directive in RQ2; YAML rules + [SLOT] masking + regex sanitizer in RQ3; slot ledger + persona trade-off policy in RQ4) rather than full sub-component ablations. Fourth, RQ4 uses cooperative LLM user simulators (validated for within-

simulation persona fidelity at 95.6%, $\kappa = .933$) and a 5-turn protocol floor ($M = 5.00$, $SD = 0.00$ in two cells).

Impact of Limitations

These limitations bound internal interpretation and external generalizability in specific, quantifiable ways: (a) continuous cluster boundaries mean ~25.2% of boundary shoppers sit near Voronoi hyperplanes where hard single-cluster assignment can misroute mixed behaviors; (b) bundled treatments establish the efficacy of each integrated stage but prevent attributing exact variance shares to exemplars versus directives without follow-up ablation; and (c) cooperative simulators and the 5-turn floor mean that the 29.5%–42.7% turn reductions represent an upper-bound controlled benchmark relative to noisy production traffic where users abandon chats due to exogenous shocks (such as leave denial or competitor browsing).

Future Improvements

Immediate methodological improvements include: (1) executing orthogonal sub-component ablations, expanding RQ2 to a 5-arm design crossing Exemplar Source (None / Static / Archetype-Partitioned) with Pacing Directive (Off / On) at $N \geq 60$, separating pre-generation [SLOT] masking from post-generation regex sanitization in RQ3, and crossing the Slot Ledger (Off / On) with Persona Policy (Generic / Adaptive) in a 2 × 2 × 3 RQ4 experiment; (2) introducing **hierarchical context × behavior clustering ($C \times K = 24$–36 micro-clusters)** over situational trip stakes (*party_type*, *budget_stakes*, *lead_window*) using top-15 parent FAISS retrieval with contextual metadata back-off; and (3) replacing hard K-Means routing with **soft-mixture exemplar quotas** and **turn-by-turn Bayesian posterior tracking** $P(z_t \mid u_{1..t})$ to handle boundary users and mid-chat price shocks.

Future Scope

For broader doctoral and enterprise extension across real-world tour operators, where customers rarely book in a single chat but instead engage across **4–5 chats and calls under a shared CRM *Opportunity_ID*** and may abandon final payment for exogenous reasons, future research will extend this framework in four directions: (1) replacing noisy single-chat *Turns-to-Booking* with controllable multi-session metrics, namely **Turns-to-Lead (TTL)** on Call 1, per-session **Turns-to-Quotation (TTQ$_s$)** across Calls 2–5, and **Cross-Session Excess Turns (CSET = $\Sigma_s[T_s - T_s^{\min}]$)** modeled via linear mixed-effects models and right-censored Markov transitions across persistent *Cold* → *Warm* → *Hot* lead states (pre-loading prior-call slot ledgers for 0.0% cross-session RER); (2) expanding FAISS into a **3 × 6 (Archetype × Call Stages *a–f*: Opening, Mandatory Slots, Positioning, Negotiation, Compliance, Closure)** lookup matrix; (3) indexing **200+ regional packages by Unique Selling Propositions (USPs)** queried via an information-gain question tree (*seniors, children, breakfast, sightseeing tempo, occasion*); and (4) pairing FAISS style cosine with a **Historical Turn-Economy Database (*Turn-Q DB*)** and constrained offline-to-online reinforcement learning under live multilingual OTA A/B testing.

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Appendix A

Repository Structure, Replication Guide, and Raw Source Schema

All code, data schemas, cached LLM outputs, and report builders are available at https://github.com/surnella/QM640-Dialogue-Clustering-Capstone. Table A1 outlines the repository directory structure, Table A2 lists the exact shell commands that reproduce every table and figure, and Table A3 provides Part A (raw source fields) and Part B (canonical schema fields) of the data dictionary.

Table A1

Repository Structure and Artifact Inventory

Directory or file path	Contents and analytical purpose
data/raw/	Downloaded Microsoft Frames (1,369 dialogues) and MultiWOZ 2.2 corpora (excluded from Git)
data/canonical/	Normalized two-tier Pydantic dialogue trajectories (canonical_corpus.jsonl)
data/interim/	Cleaned samples, partitions, embeddings, primary/refined archetypes, LLM annotations, and RQ1/RQ2 JSON summaries
data/final/	Cached Gemini 2.5 Flash/Pro JSONL responses, feature/router diagnostics, and scored CSV/JSON results for RQ2–RQ4
src/canonical_schema.py, src/ingestion.py	Pydantic two-tier data models and corpus parsers
src/rule_engine.py, rules/rules_enterprise.yaml	Deterministic declarative policy engine and YAML rule definitions
src/final_pipeline/	Reproducible pipeline scripts for Steps 0–5 (power, cleaning, RQ1 clustering/refined archetypes, router, RQ2–RQ4, feature importance, & 300-DPI figures)
src/report/, src/build_final_report.py	APA 7 Word/PDF document generator with automated cross-reference, word-count, and TOC pagination verification
notebooks/01_.. to 05_..	Five executed Jupyter notebooks covering EDA, RQ1, RQ2, RQ3, and RQ4
outputs/interim/, outputs/final/	All CSV summary tables and 1:1 6.5-inch 300-DPI PNG figures
final/	Submission-ready QM_640_Final_Report_Sandeep_Surnella.docx and .pdf

Table A2

End-to-End Replication Commands

Pipeline stage	Command (run from repository root with .venv active)	Outputs produced
0. Power analysis	python src/final_pipeline/step0_sample_size.py	Table 8
1. Cleaning & univariate EDA	python src/final_pipeline/step1_prepare_data.py	Tables 5, 6, E1
2. Primary RQ1 clustering	python src/final_pipeline/step2_rq1_clustering.py	Table 14; Tables D1–D4; Figure D3
2b–2i. Sensitivity & refined	python src/final_pipeline/step2g_rq1_truncation.py; step2h_rq1_ablation.py; step2i_rq1_refined_archetypes.py	Table 15; Tables C1–C2, D5–D6, E5, F1
3. Prefix routing harness	python src/final_pipeline/test_step1_harness.py	Table 15 routing diagnostics
4. Refined RQ2 experiment	python src/final_pipeline/step2_rq2_refined.py --execute	Table 16; Table E3
5. RQ3 2×2 governance exp.	python src/final_pipeline/step3_rq3_experiment.py --execute	Table 17; Tables C3, G1
6. RQ4 2×3 turn-economy sim.	python src/final_pipeline/step4_rq4_simulation.py --execute	Table 18; Tables C3, G1
7. Feature importance & figs.	python src/final_pipeline/step5_figures_and_eda.py	Tables 6, 7, 11–13, E2; Figures 1–9, D1, D2, D4, E1–E3
8. Compile Final Report	python src/build_final_report.py	final/QM_640_Final_Report_Sandeep_Surnella.docx & .pdf

Note. All random seeds are fixed at 42. Cached JSONL responses allow full offline execution without external API calls.

Table A3

Data Dictionary, Parts A and B: Raw Corpus Fields and Canonical Schema Mapping

Source / Layer	Field name	Data type (Unit)	Description and mapping role
Frames (Raw)	id, user_id, wizard_id	String (ID)	Unique dialogue ID and role-player IDs (11 users, 12 wizards)
Frames (Raw)	turns[].author, text, timestamp	String / Float (ms)	Speaker (user or wizard), raw utterance text, and Unix timestamp
Frames (Raw)	turns[].labels.acts[].name, args	List / Key–Value	Dialogue act tags (inform, request, offer, switch_frame) and slot–value pairs

Source / Layer	Field name	Data type (Unit)	Description and mapping role
Frames (Raw)	labels.wizardSurveyTaskSuccessful, userSurveyRating	Boolean / Float (1–5)	Wizard task completion flag and user satisfaction rating
MultiWOZ (Raw)	dialogue_id, services	String / List	Unique dialogue ID and active service domains (hotel, train, taxi, attraction)
MultiWOZ (Raw)	turns[].speaker, utterance, frames[].state	String / Dict	Speaker (USER or SYSTEM), utterance text, and turn-level slot state dictionary
Canonical (Tier 1)	origin_city, destination_city, accommodation_tier	String (City / Tier)	Normalized origin, destination, and hotel star rating or category
Canonical (Tier 1)	departure_date, return_date, duration_nights	String (Date) / Int (nights)	Normalized trip dates and length of stay in nights
Canonical (Tier 1)	traveler_count_adults, traveler_count_children, budget_amount	Int (persons) / Float (USD)	Party composition and numeric budget ceiling in US dollars
Canonical (Tier 2)	visa_processing_window_days, passport_expiry_months	Int (days / months)	Departure lead window (days) and remaining passport validity (months)
Canonical (Tier 2)	package_type, sharing_basis, discount_applied_percentage	String / Float (%)	Package class (FIT or Group), room occupancy, and promotional discount (%)

Appendix B

Ten-Week Project Schedule and Milestone Completion Log

Table B1 documents the completed 10-week execution schedule aligned with QM640 course milestones (referenced in the Repository & Reproducibility overview and Section 6).

Table B1

Ten-Week Project Schedule and Milestone Completion Status

Week	QM640 course topic	Completed project tasks and deliverables	Status
1	Define research problem & topic	Business & technical problem statements finalized; GitHub repository initialized; Frames & MultiWOZ 2.2 ingested	Complete
2	Synopsis draft	Four research questions, formal hypotheses, and two-tier Pydantic schema defined	Complete
3	Literature search & data collection	30 sources synthesized across 14-row relevance matrix; canonical corpus parsed (N = 2,833 analytic)	Complete
4	Objectives & hypotheses	Data quality audit, univariate/outcome/bivariate/multivariate EDA, noncentral-F power analysis, and primary RQ1 clustering	Complete
5	Research design & methodology	Interim report submitted; primary RQ2 experiment (99 calls) and RQ1 sensitivity/truncation grid (Steps 2b–2i) completed	Complete
6	Data analysis plan	Built RefinedArchetypeRouter, feature-importance & Train/CV/Test ROC-AUC diagnostics; repeated RQ2 on refined archetypes ($p < .001$)	Complete
7	Data analysis plan execution	Executed RQ3 2 × 2 factorial experiment (N = 144) and dual-LLM judge reliability audit (*gemini-2.5-pro*, κ = .917)	Complete
8	Data analysis assessment	Executed RQ4 2 × 3 multi-turn simulation (N = 180), persona fidelity validation (95.6%, κ = .933), and HC3/ART checks	Complete
9	Refinement & documentation	Regenerated 1:1 6.5-inch 300-DPI figures, confusion matrix, ROC curves, and synchronized notebooks	Complete
10	Document results & findings	Compiled, audited, and exported submission-ready Final Report (.docx and .pdf)	Complete

Appendix C

LLM Behavioral Annotation Codebook, Reliability, and Dual-Judge Validation

Table C1 defines all 16 structured fields of the LLM behavioral codebook (*UserBehaviour* in *step2e_llm_user_annotation.py*: 10 ordinal 1–5 rating scales, 3 categorical goal/tone clustering fields, and 3 audit-only fields excluded from clustering; see also Appendix B, Table B1 for milestone sequence), Table C2 reports test–retest reliability across all 16 fields and criterion validity against ground-truth Frames dialogue acts, and Table C3 summarizes the cross-model dual-judge reliability in RQ3 and simulator persona validation in RQ4.

Table C1

LLM Behavioral Annotation Codebook (All 16 Structured Schema Fields)

# & Schema field	Scale & role	Operational definition and scale anchors
1. destination_flexibility	Ordinal 1–5 (Cluster)	1 = insists on one fixed destination; 5 = open to any destination recommendation
2. date_flexibility	Ordinal 1–5 (Cluster)	1 = fixed travel dates; 5 = wide date window or completely flexible timing
3. exploration_breadth	Ordinal 1–5 (Cluster)	1 = evaluates a single option; 5 = actively compares multiple destinations, hotels, or dates
4. decisiveness	Ordinal 1–5 (Cluster)	1 = hesitant or leaves undecided; 5 = commits quickly and firmly once a matching offer is shown
5. detail_orientation	Ordinal 1–5 (Cluster)	1 = asks no package details; 5 = probes amenities, star ratings, flight times, and inclusions
6. revision_tendency	Ordinal 1–5 (Cluster)	1 = never modifies initial constraints; 5 = repeatedly shifts dates, destinations, or budgets
7. question_asking	Ordinal 1–5 (Cluster)	1 = passive responder; 5 = asks frequent clarifying and comparison questions
8. politeness	Ordinal 1–5 (Cluster)	1 = curt or abrupt; 5 = consistently courteous and appreciative
9. price_sensitivity	Ordinal 1–5 (Cluster)	1 = price never mentioned or irrelevant; 5 = price/budget dominates every decision
10. information_upfront	Ordinal 1–5 (Cluster)	1 = reveals constraints gradually across turns; 5 = states nearly all trip constraints in opening message
11. primary_goal	Categorical (Cluster)	6 classes: explore_destinations, price_shopping, specific_trip_booking, compare_options, information_only, other

# & Schema field	Scale & role	Operational definition and scale anchors
12. secondary_goals	Multi-label (Cluster)	Multi-hot subset of the 6 goal classes above clearly present in user utterances
13. emotional_tone	Categorical (Cluster)	4 classes: positive, neutral, impatient, or frustrated
14. outcome	Categorical (Audit)	4 classes: booked, no_suitable_option, undecided_or_left, unclear (excluded from clustering input)
15. party_type	Categorical (Audit)	6 classes: solo, couple, family, group, business, unknown (excluded from clustering input)
16. unusual_conversation	Boolean (Audit)	True only if off-task or anomalous request (excluded from clustering input)

Note. Fields 1–13 form the clustering feature matrix (*llm_matrix*: 10 ordinal + 6 primary_goal + 6 secondary_goals + 4 emotional_tone = 26 numeric columns). Fields 14–16 are retained strictly for post-clustering validation.

Table C2

Test–Retest Reliability and Criterion Validity Across All 16 LLM Behavioral Schema Fields

Schema field (#1–16)	Test–retest κ	Ground-truth Frames act criterion or field role	Spearman ρ ($p < .001$)
1. Destination flexibility	.96	N/A	N/A
2. Date flexibility	.89	N/A	N/A
3. Exploration breadth	.95	Frame switches per user turn	.30
		Comparison requests per user turn	.26
4. Decisiveness	.91	N/A	N/A
5. Detail orientation	.89	Distinct preference slots mentioned	.45
6. Revision tendency	.89	Slot-value changes in user acts	.73
7. Question asking	.90	Request acts per user turn	.65
8. Politeness	.91	N/A	N/A
9. Price sensitivity	.95	Numeric budget stated	.28
10. Information given up front	.87	Share of slots stated in first user turn	.32
11. Primary goal (6 classes)	.78	Categorical Cohen's κ (clustering)	N/A
12. Secondary goals (multi-label)	.75	Binary presence Cohen's κ (clustering)	N/A
13. Emotional tone (4 classes)	.88	Categorical Cohen's κ (clustering)	N/A
14. Session outcome (4 classes)	.98	Categorical Cohen's κ (audit-only)	N/A
15. Party type (6 classes)	.93	Categorical Cohen's κ (audit-only)	N/A

Schema field (#1–16)	Test–retest κ	Ground-truth Frames act criterion or field role	Spearman ρ ($p < .001$)
16. Unusual conversation (boolean)	.80	Boolean Cohen's κ (99.0% raw agreement, audit-only)	N/A

Note. Test–retest subsample $n = 100$ (quadratic-weighted κ for ordinal Fields 1–10; Cohen's κ for Fields 11–16); criterion validity $N = 1{,}334$ Frames dialogues.

Table C3

Cross-Model Dual-Judge Reliability (RQ3) and Simulator Persona Fidelity Validation (RQ4)

Validation audit	Sample *n*	Evaluator models compared	Raw agreement / Accuracy	Cohen's κ (PABAK)	Additional validity evidence
RQ3 Policy Compliance Dual-Judge Audit	48	*gemini-2.5-flash* vs. *gemini-2.5-pro*	95.8%	.917 (.917)	Flash vs. deterministic regex audit: 89.6% agreement, κ = .791
RQ4 Simulator Persona Fidelity Audit	45	Intended Archetype vs. Blind *gemini-2.5-pro* Judge	95.6% (vs. 33.3% chance)	.933 ($p < .001$)	Mean ratings (Decisiveness / Exploration / Rigidity): DB = 5.0/1.0/1.5; OE = 3.9/4.3/2.0; RCS = 4.5/1.1/4.3

Appendix D

RQ1 Sensitivity Grid, Ablation Diagnostics, and Primary Archetype Profiles

Tables D1, D2, D3, D4, D5, and D6 and Figures D1, D2, D3, and D4 provide the complete diagnostic tables and visual plots supporting the RQ1 primary clustering benchmark, Tukey/Games–Howell post hoc algorithm contrasts, primary whole-dialogue archetype profiles, cross-corpus centroid projection, and the representation × truncation sensitivity and ablation analysis.

Table D1

Choice of Number of Clusters k on the Golden Partition Across Initial MiniLM and Final 46-d Blocks (M = 9) Representations

Representation & *k*	Cosine Silhouette	Davies–Bouldin index	Inertia / Bootstrap ARI	Smallest cluster (%) & diagnostic
Panel A (MiniLM-L6-v2 whole): *k* = 2	.206	3.136	Inertia = 193.1	35.5%
Panel A (MiniLM-L6-v2 whole): *k* = 3	.141	3.244	Inertia = 184.6	24.1%
Panel A (MiniLM-L6-v2 whole): *k* = 4	.142	2.997	Inertia = 178.1	17.7%
Panel A (MiniLM-L6-v2 whole): *k* = 5	.148	2.950	Inertia = 173.6	8.7%
Panel A (MiniLM-L6-v2 whole): *k* = 6	.077	3.118	Inertia = 169.7	8.4%
Panel B (46-d Blocks, *M* = 9): *k* = 2	.112 (cal. .044)	2.796	ARI = .949 (96.2% core)	40.4%: Collapses Decisive & Rigid into 1 cluster
Panel B (46-d Blocks, *M* = 9): *k* = 3	.094 (cal. .033)	2.647	ARI = .888 (93.1% core)	25.5%: Adopted winner (lowest DB, ARI >= .60, min >= 20%)
Panel B (46-d Blocks, *M* = 9): *k* = 4	.066 (cal. .015)	2.756	ARI = .589 (55.8% core)	12.5%: Fails stability (ARI < .60) & min size (< 20%)

Representation & *k*	Cosine Silhouette	Davies–Bouldin index	Inertia / Bootstrap ARI	Smallest cluster (%) & diagnostic
Panel B (46-d Blocks, *M* = 9): *k* = 5	.069 (cal. .026)	2.784	ARI = .606 (61.1% core)	11.4%: Fails stability (ARI < .60) & min size (< 20%)
Panel B (46-d Blocks, *M* = 9): *k* = 6	.065 (cal. .019)	2.729	ARI = .631 (53.2% core)	11.3%: Fragmented min cluster (11.3% < 20% floor)

Note. On the final 46-d *Blocks* (*M* = 9) representation (Panel B), *k* = 3 achieves the lowest (best) Davies–Bouldin index (2.647), high bootstrap stability (ARI = .888 in the truncation grid; .869 in the 6-config ablation), and balanced cluster sizes (≥ 25.5%).

Table D2

Pairwise Post Hoc Comparisons of Mean Silhouette Across Clustering Algorithms

Algorithm comparison	Mean difference	95% CI (Tukey HSD)	*p* (Tukey HSD)	*p* (Games–Howell)
K-Means − Agglomerative	.029	[.013, .046]	< .001	< .001
HDBSCAN − Agglomerative	.093	[.076, .110]	< .001	< .001
HDBSCAN − K-Means	.063	[.047, .080]	< .001	< .001

Table D3

Behavioral Profiles of the Three Primary Whole-Dialogue Archetypes (Golden Partition)

Feature	Direct Transactional	Constraint Negotiator	Guided Explorer
n (%)	240 (25.7%)	225 (24.1%)	468 (50.2%)
Turns, *M* (*SD*)	9.92 (5.02)	15.64 (6.84)	17.19 (6.10)
Words per user / agent turn	10.60 / 11.67	12.94 / 18.15	9.63 / 15.71
Slot-value revisions / Hesitations	6.9 / 0.7	20.3 / 1.1	16.7 / 1.1
Agent package offers	0.84	4.28	3.73
Budget stated (%) / Task success (%)	63.7% / 92.1%	70.7% / 94.2%	75.4% / 95.3%

Table D4

Primary Archetype Assignment and Centroid Similarity Across Partitions and MultiWOZ 2.2

Sample	*N*	Centroid cosine *M* (*SD*)	Direct Transactional (%)	Constraint Negotiator (%)	Guided Explorer (%)
Frames golden	933	.895 (.042)	25.7	24.1	50.2
Frames held-out	401	.894 (.042)	29.4	20.2	50.4
MultiWOZ 2.2	1,499	.805 (.037)	0.9	15.1	84.0

Table D5

Stability, Routing Agreement, and Outcome Association of Hybrid Archetypes by Conversation Length

User messages used (*M*)	Dialogues shortened (%)	Bootstrap ARI	Core members (%)	Smallest cluster (%)	Raw L1 κ	Blocks router κ	Outcome *V*
First 3 messages	94.7	.89	92.7	18.3	.72	.359	.16
First 5 messages	78.2	.84	77.1	20.3	.38	.535	.27
First 7 messages	53.2	.85	89.9	29.7	.25	.722	.38
First 9 messages	30.7	.89	93.1	25.5	.24	.958	.45
Whole conversation	0.0	.62	53.7	7.9	.22	N/A	.57

Note. Raw L1 κ uses unprojected 5-turn features (*step2g*); Blocks router κ projects prefix features into the exact golden *Blocks* space (*RefinedArchetypeRouter*).

Table D6

Ablation of the Refined Archetype Definition at M = 9 User Messages (k = 3)

Representation configuration	ARI (whole)	ARI (9 msgs)	Core members (%)	Outcome *V*	ARI vs. adopted
Gemini embedding + Gemini behavior profile (adopted)	.56	.87	89.2	.45	1.00
MiniLM embedding + Gemini behavior profile	.69	.85	87.7	.40	.33
Gemini behavior profile only	.73	.85	97.6	.35	.17
Gemini embedding only	.91	.91	92.0	.34	.45
MiniLM embedding only	.68	.72	72.1	.27	.14

Representation configuration	ARI (whole)	ARI (9 msgs)	Core members (%)	Outcome *V*	ARI vs. adopted
MiniLM embedding + dialogue-act features (no LLM)	.61	.76	88.0	.45	.16

Figure D1

Cluster-Number Diagnostics on Golden Partition

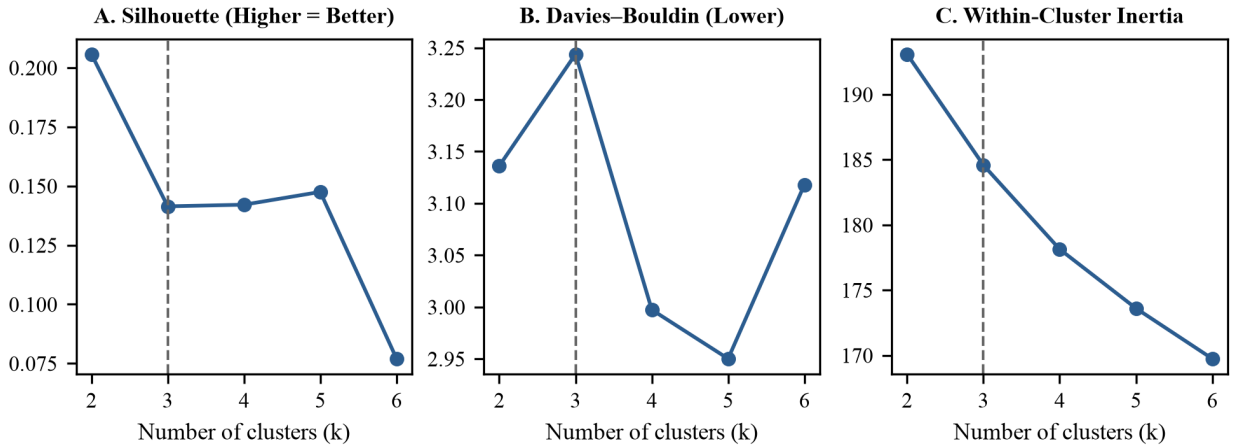


Figure D2

Silhouette Coefficients Across 159 Matched Resampling Runs by Algorithm

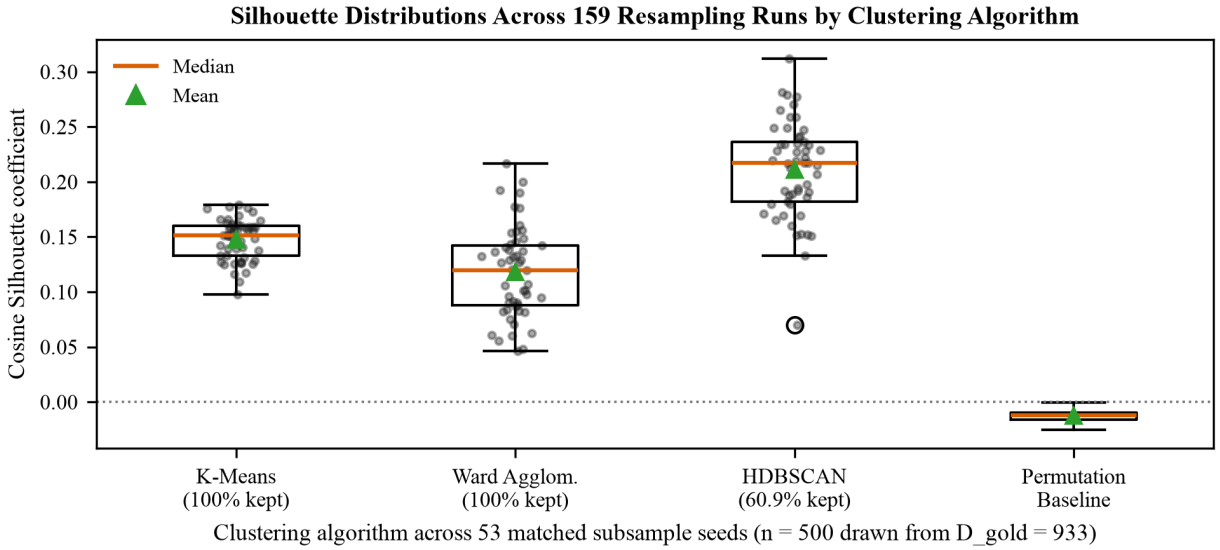


Figure D3

UMAP Projection of Frames Archetypes and MultiWOZ 2.2 External Transfer Check

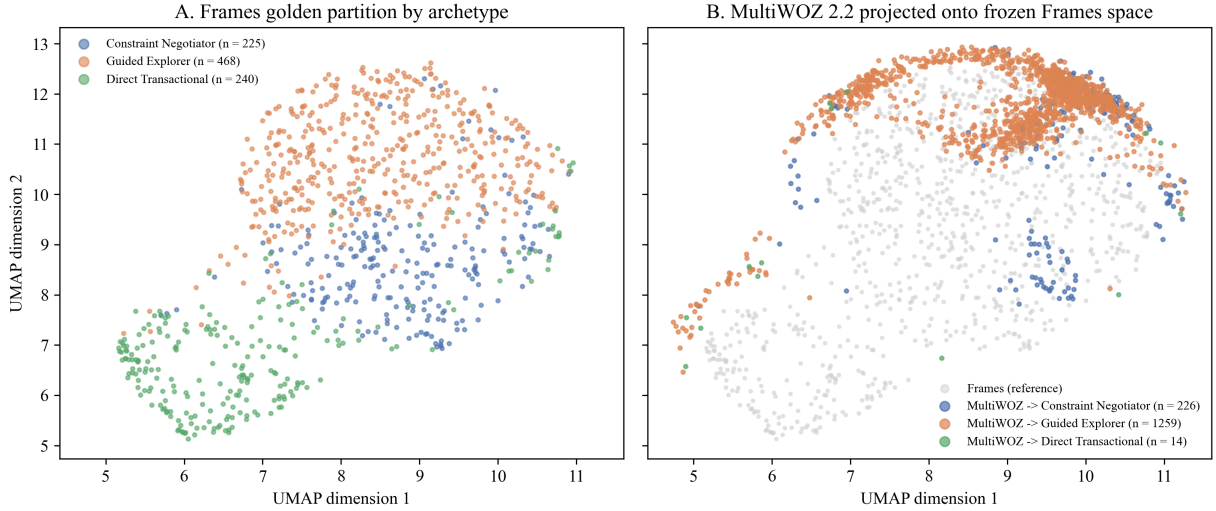
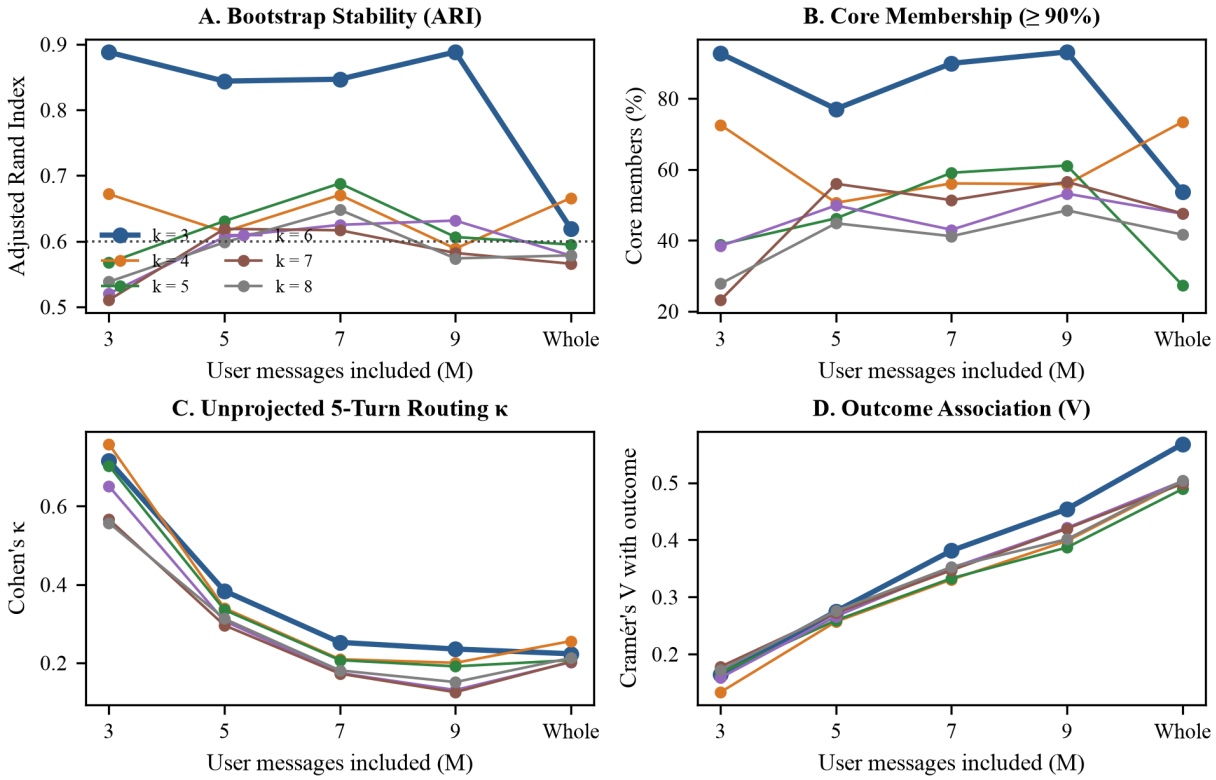


Figure D4

Hybrid Archetype Stability, Routing, and Outcome Association Across Truncation Horizons



Appendix E

Supplementary EDA, RQ2 Pairwise Contrasts, and RQ3/RQ4 Groundwork Diagnostics

Tables E1, E2, E3, E4, and E5 and Figures E1, E2, and E3 provide the supplementary slot-completeness table and plot, multivariate PCA loadings, Holm-adjusted pairwise contrasts and primary unrefined evaluation for RQ2, corpus rule coverage audit for RQ3, and human turn-economy baseline for RQ4.

Table E1

Observed Canonical Slot Completeness by Corpus

Canonical slot	Schema tier	Microsoft Frames (%)	MultiWOZ 2.2 (%)
origin city	Tier 1	99.4	70.8
destination city	Tier 1	99.0	71.4
departure date	Tier 1	84.9	68.6
return date	Tier 1	70.8	0.0
duration nights	Tier 1	61.8	27.0
traveler count adults	Tier 1	63.9	54.0
traveler count children	Tier 1	18.4	0.0
budget amount	Tier 1	72.9	24.1
accommodation tier	Tier 1	72.7	19.7
visa processing window days	Tier 2	0.0	0.0
passport expiry months	Tier 2	0.0	0.0
discount applied percentage	Tier 2	0.0	0.0

Table E2

Multivariate Principal Component Loadings of Structural and Behavioral Features (Frames)

Variable	PC1 loading (40.8% var.)	PC2 loading (17.2% var.)	PC3 loading (10.6% var.)
Total Turns	.393	.166	−.234
Slot Revisions	.424	.195	−.064
Agent Offers	.361	.292	−.078
User Words/Turn	.025	−.087	.888
Dest. Flexibility	.327	−.182	.135

Variable	PC1 loading (40.8% var.)	PC2 loading (17.2% var.)	PC3 loading (10.6% var.)
Exploration Breadth	.417	−.108	.101
Decisiveness	−.259	.503	.028
Revision Tendency	.297	−.460	.025
Question Asking	.301	.248	.027
Politeness	.089	.520	.341

Table E3

Holm-Adjusted Paired Contrasts for Refined RQ2 Outcomes (N = 33)

Outcome metric	Pairwise contrast	Mean diff.	*t*	*df*	*p* (Holm)	d_z
Refined: SBERT style cosine	Cluster-Adaptive vs. Static	.144	4.24	32	< .001	0.74
Refined: SBERT style cosine	Cluster-Adaptive vs. Zero-Shot	.067	2.36	32	.049	0.41
Refined: SBERT style cosine	Static vs. Zero-Shot	−.077	−2.20	32	.049	−0.38
Refined: Gemini style cosine	Cluster-Adaptive vs. Zero-Shot	.027	1.20	32	.713	0.21
Refined: Gemini style cosine	Static vs. Zero-Shot	.024	0.79	32	.865	0.14
Refined: Gemini style cosine	Cluster-Adaptive vs. Static	.004	0.13	32	.895	0.02
Refined: BERTScore F1	Cluster-Adaptive vs. Static	.056	1.45	32	.472	0.25
Refined: BERTScore F1	Cluster-Adaptive vs. Zero-Shot	.025	0.85	32	.804	0.15
Refined: BERTScore F1	Static vs. Zero-Shot	−.031	−0.75	32	.804	−0.13
Refined: Pacing error (words)	Cluster-Adaptive vs. Zero-Shot	−6.25	−3.49	32	.004	−0.61
Refined: Pacing error (words)	Cluster-Adaptive vs. Static	−2.46	−2.66	32	.024	−0.46
Refined: Pacing error (words)	Static vs. Zero-Shot	−3.79	−1.82	32	.078	−0.32
Refined: Response length (words)	Cluster-Adaptive vs. Static	6.82	7.00	32	< .001	1.22
Refined: Response length (words)	Static vs. Zero-Shot	−10.91	−4.29	32	< .001	−0.75
Refined: Response length (words)	Cluster-Adaptive vs. Zero-Shot	−4.09	−1.84	32	.075	−0.32

Table E4

Empirical Corpus Audit of Policy Rule Coverage in Microsoft Frames (N = 1,334)

Declarative rule	Required canonical slot	Slot observed (%)	Dialogues triggering (*n*)	Triggering (%)
pax capacity	traveler count adults	24.1	54	4.05
budget minimum	budget amount	65.1	266	19.94
visa window	visa processing window days	0.0	0	0.00
passport validity	passport expiry months	0.0	0	0.00
discount threshold	discount applied percentage	0.0	0	0.00

Table E5

Human Wizard Baseline of Turn Economy and Redundant Elicitation by Refined Archetype (Frames)

Refined user archetype	*n*	Whole-dialogue turns *M* (*SD*)	RER *M* (*SD*)	Wizard request turns *M*
Decisive Booker	595	14.56 (4.52)	.014 (.047)	1.44
Open Explorer	337	20.75 (7.07)	.016 (.049)	1.82
Rigid-Constraint Seeker	402	10.60 (5.59)	.017 (.061)	1.27

Note. Turns in Table E5 reflect total observed human-wizard turns across complete archival Frames transcripts, whereas Table 18 (RQ4) measures total conversational turns until package agreement (*[ACCEPT_BOOKING]*) in closed-loop simulation.

Figure E1

Observed Canonical Slot Completeness by Corpus

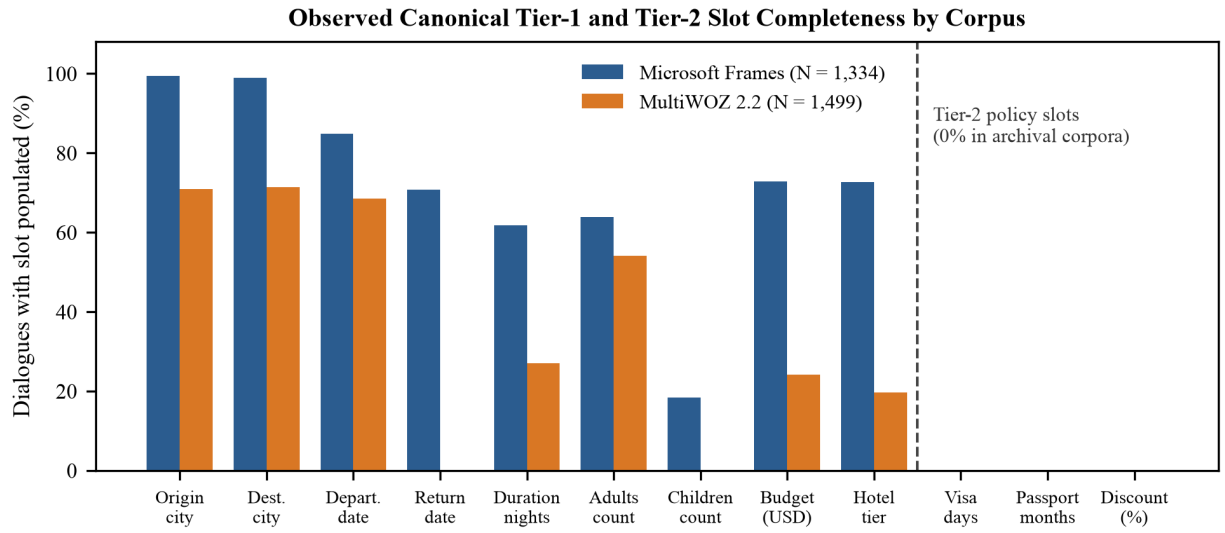


Figure E2

Primary RQ2 Results on Unrefined Whole-Dialogue Archetypes

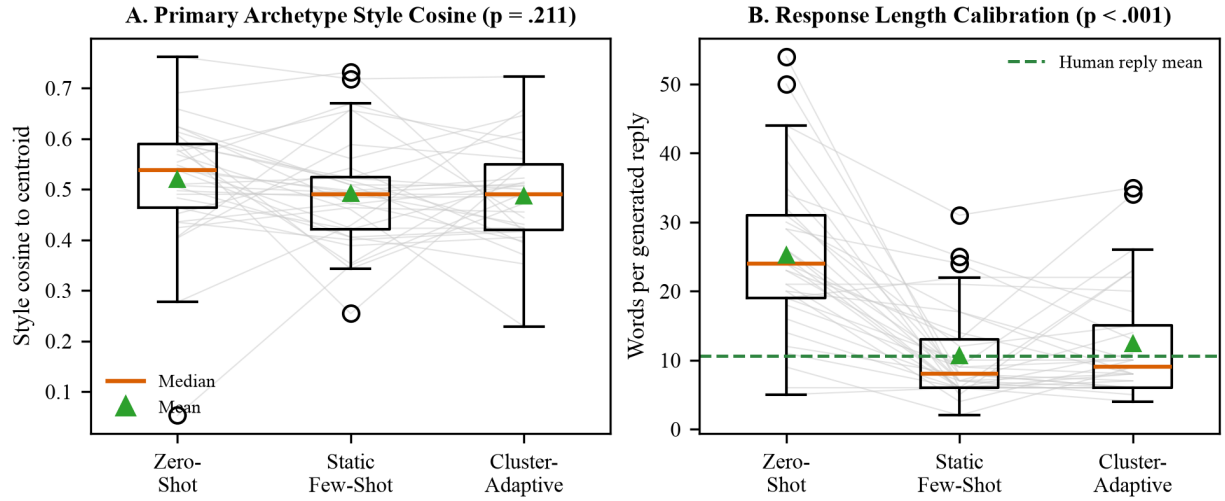
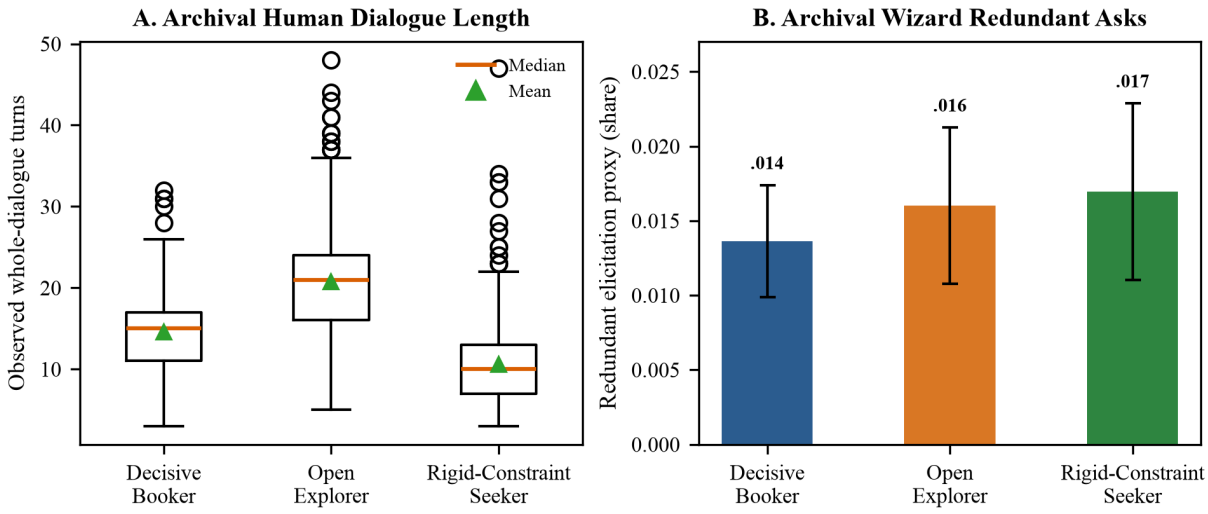


Figure E3

Human Wizard Baseline of Turn Economy and Redundant Elicitation by Refined Archetype



Appendix F

Derivation of User Archetype Cluster Names and Deterministic Centroid-Mapping Protocol

Unsupervised clustering algorithms such as *K*-Means output anonymous integer cluster identifiers (*0*, *1*, *2*) rather than semantic persona names, and those integer identifiers are invariant up to permutation across random initializations (the *label-switching problem*). To ensure that human-interpretable archetype labels are assigned reproducibly using the highest-importance behavioral features (Table 13), both the primary whole-dialogue pipeline (*src/final_pipeline/step2_rq1_clustering.py*) and the refined opening-prefix pipeline (*src/final_pipeline/step2i_rq1_refined_archetypes.py*) apply pre-declared, deterministic Python centroid-mapping rules to the golden-partition cluster means (Table F1).

Table F1

Deterministic Centroid-Mapping Rules for Primary and Refined User Archetypes

Clustering taxonomy & script	Step & deterministic Python rule on cluster centroid means	Assigned archetype name	Distinguishing golden-partition centroid signature (*N* = 933)
Primary Whole-Dialogue Archetypes (*step2_rq1_clustering.py*)	Step 1: *idxmax(turns_M)* among the 3 clusters	Guided Explorer	Longest dialogues (*M* = 19.65 turns, *n* = 187, 20.0%); highest option exploration across whole transcripts
Primary Whole-Dialogue Archetypes (*step2_rq1_clustering.py*)	Step 2: *idxmax(revisions)* among the remaining 2 clusters	Constraint Negotiator	Highest slot-revision count (*M* = 1.98 revisions, *M* = 15.80 turns, *n* = 351, 37.6%)
Primary Whole-Dialogue Archetypes (*step2_rq1_clustering.py*)	Step 3: Remaining cluster ID	Direct Transactional	Shortest whole-dialogue sessions (*M* = 11.98 turns, *n* = 395, 42.3%); lowest revision count
Refined Opening-Prefix Archetypes (*step2i_rq1_refined_archetypes.py*)	Step 1: *idxmax(destination_flexibility)* across the 3 *Blocks* clusters (*M* = 9)	Open Explorer	Highest destination flexibility (3.49/5.0) and exploration breadth (2.96/5.0); longest dialogues (20.78 turns; *n* = 238, 25.5%)

Clustering taxonomy & script	Step & deterministic Python rule on cluster centroid means	Assigned archetype name	Distinguishing golden-partition centroid signature ($N = 933$)
Refined Opening-Prefix Archetypes (*step2i_rq1_refined_archetypes.py*)	Step 2: *idxmax(decisiveness)* among the remaining 2 clusters	Decisive Booker	Highest decisiveness (3.97/5.0), politeness (4.26/5.0), and specific trip booking goal (99.3%; $n = 420$, 45.0%)
Refined Opening-Prefix Archetypes (*step2i_rq1_refined_archetypes.py*)	Step 3: Remaining cluster ID	Rigid-Constraint Seeker	Lowest destination/date flexibility (2.20/2.18) and detail orientation (1.46/5.0); highest non-booking rate under rigid constraints (68.0% no suitable option; $n = 275$, 29.5%)

Note. All centroid-mapping rules are evaluated strictly on golden-partition means (D_{gold}, $n = 933$) after fitting *K*-Means ($k = 3$, seed = 42).

Appendix G

Role, Architecture, and Validation Scope of Judge and Annotator LLMs

All primary hypothesis tests (H_{01}–H_{04}) are anchored in deterministic mathematical and programmatic Python verification metrics rather than subjective LLM preference scores: RQ1 uses mathematical Silhouette, Davies–Bouldin, ARI, Cohen's κ, and ROC-AUC; RQ2 uses deterministic non-Gemini encoders (*all-MiniLM-L6-v2* cosine, BERTScore F_1, and exact word-count error); RQ3 computes entity leakage (ELR) and Slot F_1 via exact Python set verifiers (*src/final_pipeline/metrics.py*); and RQ4 computes turn economy (TGC and RER) via deterministic turn counters and regex slot-ledger matching.

LLMs serve two bounded functions (Table G1): (a) in RQ1, *gemini-2.5-flash* (temperature = 0) extracts the 16-field *UserBehaviour* codebook prior to clustering (validated via test–retest κ_w = .87–.96 and Frames act correlations ρ = .30–.73; Table C2); and (b) in RQ3 and RQ4, *gemini-2.5-pro* acts strictly as an independent cross-verification gatekeeper (95.8% agreement, κ = .917 with *gemini-2.5-flash* and 93.1% agreement, κ = .861 with deterministic regex in RQ3; 95.6% blind persona accuracy, κ = .933 in RQ4; Table C3).

Table G1

Functional Scope, Primary Deterministic Metrics, and Cross-Verification Role of Judge and Annotator LLMs

Research question & pipeline stage	Primary hypothesis-test metrics (Deterministic / Mathematical)	LLM annotator / Judge model & role	Cross-verification agreement & gatekeeping outcome
RQ1-Identify Stage: Opening-Prefix Behavioral Annotation (*step2e_llm_user_annotation.py*)	Cosine Silhouette, Davies–Bouldin, Bootstrap ARI (.89 / .87), feature importance (η^2, $\Delta\kappa$), and prefix κ / ROC-AUC	**Codebook Annotator (*gemini-2.5-flash*, temp = 0):** Extracts Fields 1–13 (clustering inputs) and Fields 14–16 (audit-only) before *K*-Means runs.	Test–retest κ_w = .87–.96 (mean .91) on ordinal Fields 1–10; κ = .75–.98 on Fields 11–16; Spearman ρ = .30–.73 ($p < .001$) vs. Frames acts (Table C2).
RQ2-Personalize Stage: Style Convergence & Pacing Calibration (*step2_rq2_refined.py*)	De-lexicalized SBERT style cosine (*all-MiniLM-L6-v2*), BERTScore F_1, and absolute word-count error	**No Judge LLM used:** All evaluation encoders are deterministic, non-Gemini neural and lexical metrics.	Eliminates same-family LLM evaluation bias in RQ2; RM-ANOVA and Friedman tests confirm $p <$ .001 on SBERT style convergence.
RQ3-Govern Stage: Policy Compliance Dual-Judge Audit (*step3_rq3_experiment.py*)	Deterministic regex rule verifier (*det_pass*), exact Exemplar Entity Leakage Rate (ELR), and deterministic Slot Grounding F_1	**Primary Judge (*gemini-2.5-flash*) + Independent Validator (*gemini-2.5-pro*):** Cross-verifies enforcement of 5 policy rules on stratified audit subsample (n = 48).	Guided *pilot_v1* → *pilot_v2* prompt ordering fix; final dual-judge agreement = 95.8% (κ = .917, PABAK = .917) and 93.1% (κ = .861) vs. regex (Table C3).
RQ4-Optimize Stage: Blind Simulator Persona Fidelity Audit (*step4_rq4_simulation.py*)	Deterministic turn counter to agreed package quotation (*total_turns*, *log_turns*), regex Redundant Elicitation Rate (RER), and completion rate	**Independent Blind Validator (*gemini-2.5-pro*):** Reads simulated customer transcripts (n = 45, 15/archetype) blinded to true labels; contributes zero metrics to RQ4 ANOVA.	Classified 43/45 simulated users into their intended archetype (95.6% accuracy vs. 33.3% chance, Cohen's κ = .933, $p < .001$; Table C3).

Note. All LLM annotation and judge calls use deterministic decoding (temperature = 0.0, Pydantic JSON schema enforcement) and are cached in *data/interim/* and *data/final/*.